International Journal of Remote Sensing

ISSN: 0143-1161 (Print) 1366-5901 (Online) Journal homepage: www.tandfonline.com/journals/tres20

Hyperspectral change detection: an experimental comparative study

Mahdi Hasanlou & Seyd Teymoor Seydi

To cite this article: Mahdi Hasanlou & Seyd Teymoor Seydi (2018) Hyperspectral change detection: an experimental comparative study, International Journal of Remote Sensing, 39:20, 7029-7083, DOI: 10.1080/01431161.2018.1466079

To link to this article: https://doi.org/10.1080/01431161.2018.1466079

Published online: 26 Apr 2018.

INTERNATIONAL JOURNAL OF REMOTE SENSING
2018, VOL. 39, NO. 20, 7029–7083
https://doi.org/10.1080/01431161.2018.1466079

Taylor & Francis
Taylor & Francis Group

Check for updates

Hyperspectral change detection: an experimental comparative study

Mahdi Hasanlou and Seyd Teymoor Seydi

School of Surveying and Geospatial Engineering, College of Engineering, University of Tehran, Tehran, Iran

ABSTRACT

The Earth's surface is constantly changing due to variations originating from the increasing human population. In the last decade, numerous methods were presented in the literature for change detection using multispectral image data. Owing to the increasing availability of hyperspectral images, these methods are now being applied to hyperspectral images. The main objective of this study is to present different change detection methods in hyperspectral imagery. Numerous algorithms (more than 43 algorithms) have been proposed for change detection in hyperspectral imagery over the last decade. In this work, we provide a comparative review of these algorithms through experimental results. We place the algorithms in five major groups: (1) match-based, (2) transformation-based, (3) direct classification-based, (4) post-classification-based, and (5) hybrid-based. We evaluate and compare the performances of all five groups using two real-world data sets of multi-temporal hyperspectral imagery. This comparative study investigates the advantages and disadvantages of the effects of preprocessing steps in the efficiency of the hyperspectral change detection (HSCD) methods. These preprocessing steps are considered in four scenarios, including: (1) considering only spatial or geometric correction without noise reduction and spectral correction; (2) spatial, atmospheric, and radiometric corrections without noise reduction; (3) spatial correction and noise reduction without atmospheric and radiometric corrections; and (4) spatial, atmospheric, and radiometric correction with noise reduction. The empirical results, followed by a summary of the pros and cons of each algorithm, aim to help researchers select the procedures with the best characteristics for HSCD applications.

ARTICLE HISTORY
Received 10 January 2017
Accepted 5 April 2018

1. Introduction

The Earth's surface undergoes constant changes over time due to influences originating from the increasing human population (Alberti et al. 2003). Hence, an analysis of the changes is necessary for the better management of natural resources and preventing disasters (Kumar and Garg 2013). Remote sensing (RS) is a tool that provides valuable information about the environment at the lowest cost and time in a continuous manner while covering wide areas (Chen, Kumar, and Faghmous 2015). During the recent years,

CONTACT Mahdi Hasanlou hasanlou@ut.ac.ir School of Surveying and Geospatial Engineering, College of Engineering, University of Tehran, Tehran, Iran

incorporation of multi-temporal image data sets with high spectral, spatial, and temporal resolutions has been widely employed by researchers due to the development of RS systems (Bovolo and Bruzzone 2015; Liu et al. 2016; Vongsy 2007; Pieper et al. 2015b).

In addition, there has been a huge increase in analysing hyperspectral images in many of the applications in different fields. The hyperspectral sensors can be mounted on two platforms: airborne (e.g. Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) and Hyperspectral Digital Imagery Collection Experiment (HYDICE)) and space-borne (e.g. Hyperion, Compact High Resolution Imaging Spectrometer (CHRIS), and Infrared Atmospheric Sounding Interferometer (IASI)). More importantly, the series of space-borne sensors (e.g. Environmental Mapping and Analysis Program (EnMAP), precursore iperspettrale della missione applicative (PRISMA), and Hyperspectral and Infrared Imager (HypsIRI)) will be launched on a schedule that will increase the availability of hyperspectral imagery with improvement in data quality (Liu 2015). This variety of platforms has created diversity in the spatial resolution depending on hyperspectral image application. The traditional multispectral sensors, in a single observation, collect data in specific wavelengths in a few spectral bands of the electromagnetic spectrum (Ferrato and Wayne Forsythe 2012; Liu 2015). The less number of spectral bands is a primary disadvantage to multispectral sensors. Conversely, hyperspectral imagery collects data in narrow wavelengths, i.e. the spectral range from the visible to shortwave infrared section of the electromagnetic spectrum. Thus, the high spectral resolution provides a higher identification power between objects as well as more details, resulting in a higher accuracy (Smith 2012; Shippert 2003; Khanday and Kumar 2016; Omo-Irabor 2016; Govender et al. 2008). In addition, it provides extensive analyses of the Earth's surface features that would be limited to coarser bandwidths collected by multispectral sensors (Ferrato and Wayne Forsythe 2012). Thanks to the fine spectral resolution, the hyperspectral imagery offers more abundant and more detailed information on spectral changes in multi-temporal scenes than the multispectral images, which can improve the change detection performance (Wu, Bo, and Zhang 2013). Figure 1 shows the differences between multispectral and hyperspectral data in three-dimensional space.

Therefore, multispectral sensors can also be used in many applications, such as classification and change detection, but hyperspectral sensors have the potential to provide better performances. One of the applications of RS is in change detection that processes differences between an object at different times for the same area (Singh 1989). The change detection should be applied in the following six main steps using an RS data set (Weng 2011; Lu, Guiying, and Moran 2014).

(1) Description of the change detection problems that it contains, defining the problems, the objectives, time, geographic location and the extent of the study area, RS system considerations, and the accuracy needed.
(2) Selected efficient RS data that this selection is based on are the characteristics of RS data (spatial, spectral, radiometric, and temporal resolution), the environment considerations (atmospheric, moisture conditions, cloud cover), and the characteristics of the landscape (complexity and a real extent of the case).
(3) Image preprocessing, applied in two main categories: the geometric correction based on relative (image to image) and absolute (ground control point) correction, and the spectral correction that contains, de-noising, radiometric correction,

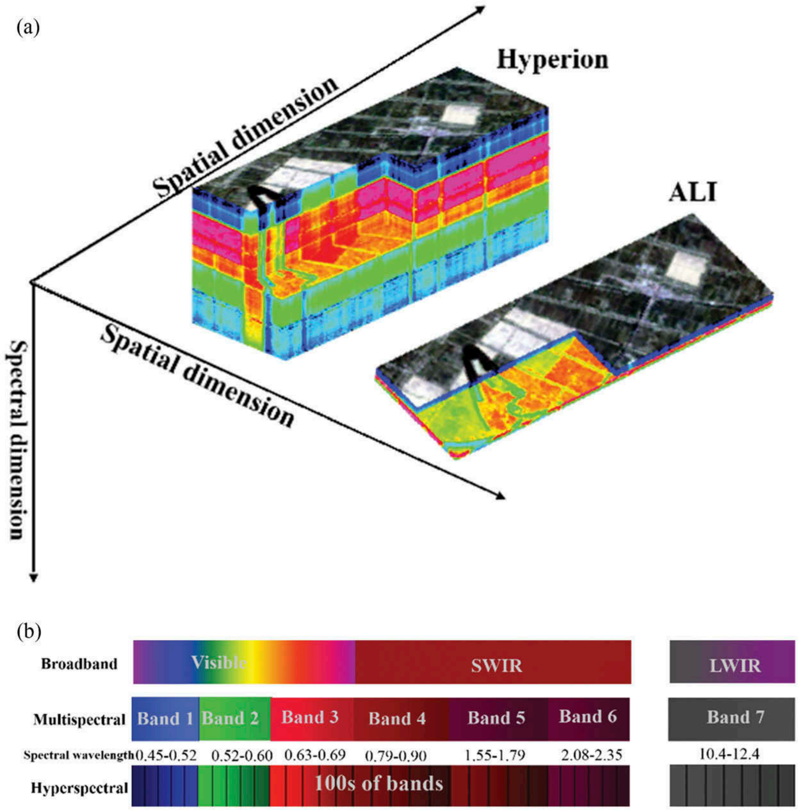


Figure 1. Comparison and contrast of hyperspectral and multispectral data sets, (a) spectral resolution of the hyperspectral (Hyperion) and multispectral (ALI) image data and (b) diagram of differences between multispectral and hyperspectral imaging (Imaging Spectroscopy 2018).

atmospheric correction (relative or absolute such as FLAASH and ACORN (Kawishwar 2007)), and the topographic corrections that need to be applied to many areas such as mountains or bumpy areas.

(4) Image processing: one of the important steps in change detection is the selection of efficient RS variables. These variables contain different features such as intrinsic RS data, per-pixel-based variables such as image transformation, thematic variables such as segmentation, spatial features such as texture images, and subpixel variables from unmixing processing such as those of the spectral mixture analysis.
(5) Selection of an appropriate change detection of algorithms is an important part of change detection. Owing to the diversity of change detection methods, it is necessary to understand the characteristics of change detection methods. If necessary, a comparison of change detection methods might be needed.
(6) Evaluation of results: accuracy assessment is a challenge in change detection. The main reason for this challenge is the difficulty in collecting reliable reference data between different temporal periods.

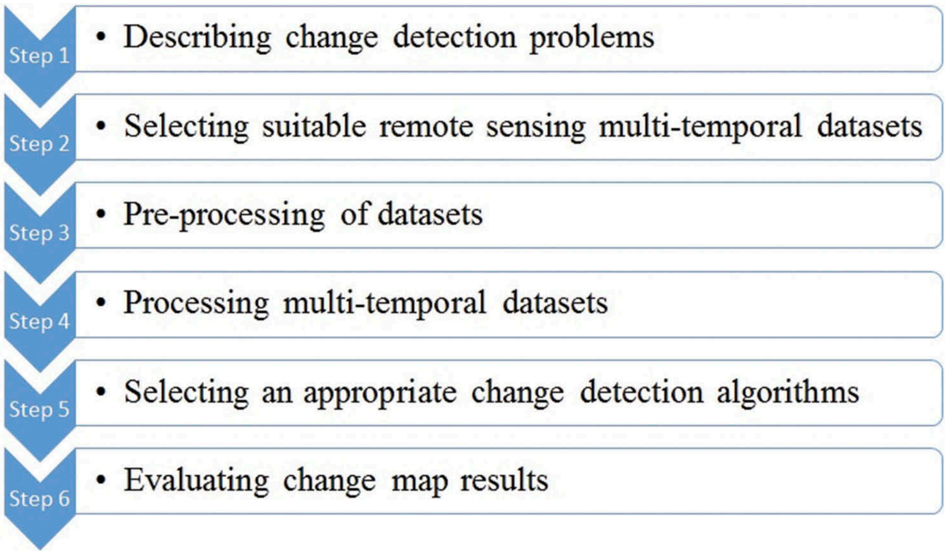


Figure 2. The main steps for change detection using remote sensing data set.

Figure 2 presents a summary of the main steps in change detection procedures using RS data sets.

The accuracy of change detection using RS data depends on some factors. These factors are as follows. (a) Change detection algorithm: it is one of the most important factors; however, to date, no change detection method has been claimed to cover all change detection problems. (b) Spatial resolution: it is an important element and a significant factor; although low-resolution data could cover a large-scale area because of monitoring more of the Earth's surface, it caused mixing of many objects, which affects the results of change detection. (c) Temporal scale: it refers to a temporal resolution that becomes constant during change detection. The variations of temporal resolution (lack of being at the same times or during the same day) effect the diurnal and seasonal Sun angles, which have a negative impact on the change detection results. (d) Image registration: one of the important steps in change detection is preprocessing. Due to lack of good registration, detected changes derived from different geographic locations of objects may cause a problem. The two main factors affect the registration, the structure of the geographic object and the spatial resolution. (e) Spectral correction: it is the difference of reflectance in multi-temporal images derived from the system malfunction of collecting data, and atmospheric attenuation of scattering and absorbing in the atmosphere (İlsever and Unsalan 2012).

The hyperspectral imagery needs special processing techniques due to various reasons such as high dimensionality and existence of significant noise (Liu 2015). In this regard, developing these specific techniques is inevitable for extracting changes and other applications of the hyperspectral imagery. Owing to the existence of different types of methods for hyperspectral change detection (HSCD), there are big challenges in incorporating these change extraction methods, and many of the factors affected the accuracy of change detection. In fact, each of these methods has its own special properties for extracting change information, which must be focused on in detail, including the accuracy of change detection, false alarm rate, 'from-to' information, atmospheric condition effect, sensitivity to noise, and complexity of procedures. Hence, it is necessary to survey different aspects of HSCD algorithms (Liu 2015; Liu et al. 2016; Seydi and Hasanlou 2016). This article focuses on the above-mentioned issues in HSCD and aims to detect, evaluate, and compare the advantages and disadvantages of each of them by implementing the different experimental algorithms.

In the last decade, several studies were conducted on different change detection methods. Vongsy (2007) studied five HSCD methods, including principal component analysis (PCA), image ratio (IR), image differencing (ID), linear chrono-chrome (CC), maximum likelihood correlation coefficient, and kernel dissimilarity measurement. The incorporated data set for that work included 33 spectral bands acquired from the rooftop of the buildings of Wright State University, and accuracy evaluation was performed using receiver operating characteristic (ROC) curve and processing time criteria. Adar et al. (2011) reviewed four HSCD methods – PCA, Linear Chrono-Chrome (LCC), Spectral Angle Mapper (SAM), and covariance equalization (CE). In addition, these algorithms were applied to the HyMap hyperspectral data set. Kumari, Gupta, and Sharma (2014) studied the common methods for HSCD, including PCA, ID, IR, linear discriminant analysis (LDA), and supervised and unsupervised classification-based methods and cited their advantages and disadvantages. Dhawan and Narula (2015) explored a number of HSCD methods, including independent component analysis (ICA), ID, IR, PCA, and change vector analysis (CVA). This work also lists the advantages and disadvantages of unsupervised and supervised post-classification-based change detection methods. Pieper et al. (2015a) first used transformation as a predictor to maximize the overall correlation between the images, and then applied an anomalous change detector (ACD). These transformations consist of canonical correlation analysis (CCA), modified CCA (MCCA), CE, PCA, and singular value decomposition (SVD), although the mentioned paper uses common transformation-based change detection. In another related work, Kamboj and Ahuja (2016) studied different types of methods, including sub-space-based (SSB), ICA, PCA, CVA, image regression, and univariate change detection. This research considered features, mechanism, and a basic principle for each of the methods. Although this work covers a range of common methods, it misses the experimentation on the hyperspectral data set.

In the previous part, we briefly overviewed the related work in HSCD methods. Most of the mentioned papers had a series of shortcomings, including the limitation in using common methods and lack of category methods. In other words, each paper used only a few methods for evaluation, irrespective of whether they implemented and experimented with the methods or only used small-area data sets. These drawbacks are due to the high dimensionality of hyperspectral imagery, noise effects, striping, and atmospheric conditions. Therefore, utilizing specific methods for extracting change information is inevitable (Liu et al. 2016; Hasanlou and Seydi 2016). In this regard, the main and primary focus of this article is dedicated to testing and evaluating different HSCD methods in different situations for achieving a better analysis of the incorporated methods. There are different criteria for evaluating each method, including accuracy, false alarm rates, complexity, 'from-to' information, atmospheric condition effect, and noise effects (Seydi and Hasanlou 2018; Shah-Hosseini, Homayouni, and Safari 2015; Liu et al. 2016).

Table 1. Different HSCD method groups implemented in this study.

Group	Name of the group
G1	Match-Based
G2	Transformation-Based
G3	Direct Classification -Based
G4	Post Classification-Based
G5	Hybrid-Based

The results of change detection methods could be considered in two main categories: (1) binary change detection and (2) multiple change detection. The binary change map does not investigate information about the nature of changes, whereas the multiple change map provides more details about the changes (Liu et al. 2016). This study implements and tests the common HSCD methods, which are divided into five main groups (Table 1). The first group (G1) includes match-based approaches. The match-based approaches contain two subgroups – similarity-based and distance-based – which aim to measure the difference between the data in all bands using math operator (Choi, Cha, and Tappert 2010; Simon Adar, Shkolnisky, and Dor 2012; Seydi and Hasanlou 2017). The second group (G2) includes transformation-based procedures that use a transformation technique in order to map the multivariate data to a new set of components based on first-order or higher-order statics (Vongsy 2007; Pieper et al. 2015b). The third group (G3) contains the direct classification-based procedures. This group of methods use only one classifier to classify multi-date data sets, where the statistical features of the changed pixels significantly vary from the original pixels (Hussain et al. 2013; Liu et al. 2016). The fourth group (G4) contains post-classification-based procedures, where classification approaches are applied to date t_1 and date t_2; the two classified maps are then compared on a pixel-by-pixel basis in order to produce the final change map. The fifth group (G5) includes hybrid-based procedures (Hussain et al. 2013). The hybrid-based groups combine a couple of previous methods to achieve new automatic or unsupervised methods.

The main purpose of this article is to evaluate a wide range of HSCD methods and create a taxonomy via placing each HSCD method in one of the five mentioned groups based on their properties and then applying those methods to real-world multi-date hyperspectral data sets to evaluate their performance. The evaluation procedures include accuracy, atmospheric condition effects, false alarm rate, and sensitivity to noise. In this regard, we designed four evaluation scenarios (Table 2): (S1) only spatial or geometric correction without noise reduction and spectral correction; (S2) spatial, atmospheric, and radiometric corrections without noise reduction; (S3) spatial correction and noise reduction without atmospheric and radiometric corrections; and (S4) spatial, atmospheric, and radiometric corrections with noise reduction.

Table 2. Different preprocessing scenarios for comparing HSCD methods.

Scenario	Spatial correction	Atmospheric and radiometric correction	Noise reduction
S1	Yes	No	No
S2	Yes	Yes	No
S3	Yes	No	Yes
S4	Yes	Yes	Yes

Table 3. Summary of incorporated abbreviations.

Abbreviation	Description	Abbreviation	Description
HSCD	Hyperspectral Change Detection	*N*-finder	N-finder Algorithm
S	Scenario Number	LSQ	Least Square
G	Group Number	PCA	Principal Component Analysis
ROC	Operating Characteristic Curve	MAD	Multivariate Alteration Detection
RMSE	Root Mean Square Error	IR-MAD	Iteratively Reweighted Multivariate Alteration Detection
RBF	Radial Bias Function	CE	Covariance Equalization
DN	Digital Number	CRC	Cross-Covariance
AUC	Area Under the Curve	SSB	Sub-Space-Based
κ	Kappa Coefficient	MLE	Maximum Likelihood Estimator
ID	Image Difference	FCM	fuzzy *c*-means
IR	Image Ratio	ISODATA	Iterative Self-Organizing Data Analysis
CCO	Cross-Correlation	KM	*k*-means
MD	Mahalanobis Distance	SVM	Support Vector Machine
ED	Euclidean Distance	MLC	Maximum Likelihood Classifier
CEN	Census Native	MiD	Minimum Distance
CEC	Census Cross	TCD	Target-Based change Detection
ICA	Independent Component Analysis	ICA-ID	ICA-Image Difference
CCA	Canonical Correlation Analysis	C^2VA	Compressed change vector analysis
CCA-AD	Canonical Correlation Analysis-Anomaly Detector	MSU	Multi-temporal spectral unmixing
CE-AD	Cross Equalization-Anomaly Detector	SCRC	Segment Cross-Covariance
M-CCA-AD	Modified Canonical Correlation Analysis-Anomaly Detector	SPIM	Segment PCA-based IR-MAD
PCA-AD	Principal Component Analysis and Anomaly Detector	SSDM	Semi-supervised distance metric
SVD-AD	Singular Value Decomposition and Anomaly Detector	KNN	*k*-Nearest Neighbour
SISAL	Simplex Identification via Split Augmented Lagrangian	SAM	Spectral Angle Mapper

This article will (1) explain general descriptions of HSCD methods in five main groups so that the users can comprehend the change detection problems in each process; (2) investigate the potential and performance of HSCD methods by considering the system error (noise, stripping, etc.), environmental conditions, threshold selection methods, and complexity of case study; and (3) steer readers to develop new change detection methods based on discussing feedbacks for each method. All of the abbreviations presented in Table 3 have a summary table to help readers quickly retrieve the exact name of a given algorithm.

2. HSCD methods

As briefly described in the previous section, the HSCD methods are divided into five groups. These groups include the match-based, transformation-based, post-classification-based, direct-classification-based, and hybrid-based methods, which will be described in detail in this section.

2.1. *Match-based*

The match-based group (G1) has two subgroups, called similarity-based and distance-based, both of which aim to measure the difference among the data in all bands using math operators. These two subgroups are widely used in change detection. The match-based group was widely used by many researchers in the last decade. The common

Table 4. G1 HSCD methods information.

No.	Method name	Setting parameter(s)	Reference software	Reference
1	SAM	– – – – – -	MATLAB	(Adar et al. 2011)
2	ID	– – – – –	MATLAB	Rivera 2005; K. M. Vongsy 2007; K. Vongsy et al. 2007; K. Vongsy et al. 2009)
3	IR	– – – – – -	MATLAB	(K. M. Vongsy 2007)
4	CCO	– – – – –	MATLAB	(Z. Yang and Mueller 2007)
5	MD	– – – – – -	MATLAB	(Z. Yang and Mueller 2007)
6	ED	– – – – –	MATLAB	(Resta et al. 2011; Dekker et al. 2013; Carvalho Júnior et al. 2011)

distance-based methods (Table 4) are ID (Rivera 2005; Vongsy 2007; Vongsy et al. 2007; Vongsy et al. 2009), IR (Vongsy 2007), Euclidian Distance (ED) (Resta et al. 2011; Júnior et al. 2011; Dekker et al. 2013), and Mahalanobis Distance (MD) (Yang and Mueller 2007). In addition, the common similarity-based methods include SAM (Adar et al. 2011) and cross-correlation (CCO) (Yang and Mueller 2007).

2.2. *Transformation-based*

2.2.1. *General transformation-based methods*

The main contribution of this group of change detection algorithms (G2) is the use of multi-temporal data set as the input and recognition via a transformation. This transformation is based on first-order (linear) or high-order statistics (nonlinear) operator such as variance, correlation, etc. (Vongsy 2007; Pieper et al. 2015b). Then, the change map is obtained by selecting a suitable threshold on the output of the transformation algorithm. The most important methods (Table 5) in the transformation-based group are PCA (Rivera 2005; Vongsy et al. 2007; Vongsy 2007; Adar et al. 2011; Vongsy et al. 2009), ICA (Xiao Benlin et al. 2008), Multivariate Alteration Detection (MAD) (Nielsena and Müllerb 2003), Iteratively Reweighted-MAD (IR-MAD) (Nielsen 2007), Cross-Covariance (CRC) (Eismann and Meola 2008; Eismann, Meola, and Hardie 2008; Adar et al. 2011; Dekker et al. 2013), Cross Equalization (CE) (Eismann and Meola 2008; Eismann, Meola, and Hardie 2008; Adar et al. 2011), Subspace-Based (SSB) (Wu, Bo, and Zhang 2013), Census Native (CEN) (Çeşmeci et al. 2014), Census Cross (CEC), and Maximum Likelihood Estimator (MLE) (Vongsy 2007; Vongsy et al. 2007; Vongsy et al. 2009).

2.2.2. *Combining transformation-based methods with the unmixing*

The hyperspectral imagery has a high spectral resolution while these images have a low spatial resolution; thus, the existence of more than one object in a ground pixel causes the spectrum gained by one pixel of the hyperspectral image to be a mix of many of the objects (Ertürk et al. 2014). Because there are many objects in a pixel of the image, the unmixing methods try to extract fractions of a spectral signature for objects in one pixel. Consequently, we consider unmixing procedures in HSCD methods to be crucial (Keshava and Mustard 2002; Iordache, Plaza, and Bioucas-Dias 2010; Bioucas-Dias et al. 2012; Martin and Plaza 2012; Ertürk, Iordache, and Plaza 2016; Ertürk, Iordache, and Plaza 2017).

Unmixing methods are applied in two steps: in the first step, the endmember in the related objects are extracted, and in the second step, abundance fractions for each endmember are decomposed (Ertürk and Plaza 2015; Du, Wasson, and King 2005; Liu et al. 2016). In the

Table 5. G2 HSCD methods information.

No.	Method name	Setting parameter(s)	Reference software	Reference
1	CEN	– – – – – -	MATLAB	(Çeşmeci et al. 2014)
2	CEC	– – – – –	MATLAB	(Çeşmeci et al. 2014)
3	ICA	– – – – – -	MATLAB	(Xiao Benlin et al. 2016)
4	CCA-AD	– – – – –	MATLAB	(Pieper, Manolakis, Truslow, et al. 2015; Pieper, Manolakis, Cooley, et al. 2015; Eismann, Meola, and Hardie 2008; Eismann and Meola 2008)
5	CE-AD	– – – – – -	MATLAB	(Eismann, Meola, and Hardie 2008; Eismann and Meola 2008; Pieper, Manolakis, Truslow, et al. 2015; Pieper, Manolakis, Cooley, et al. 2015)
6	MCCA-AD	– – – – –	MATLAB	(Pieper, Manolakis, Truslow, et al. 2015)
7	PCA-AD	– – – – – -	MATLAB	(Pieper, Manolakis, Cooley, et al. 2015; Pieper, Manolakis, Truslow, et al. 2015)
8	SVD-AD	– – – – –	MATLAB	(Pieper, Manolakis, Truslow, et al. 2015; Pieper, Manolakis, Cooley, et al. 2015)
9	SISAL	– – – – – -	MATLAB	(Ertürk and Plaza 2015)
10	*N*-finder	– – – – –	MATLAB	(Ertürk and Plaza 2015)
11	LSQ	– – – – – -	MATLAB	(Du, Wasson, and King 2005)
12	PCA	– – – – –	MATLAB	(K. Vongsy et al. 2009; Adar et al. 2011; K. M. Vongsy 2007; Rivera 2005; K. Vongsy et al. 2007)
13	MAD	– – – – – -	MATLAB	(Nielsena and Müllerb 2003)
14	IR-MAD	Epsilon for iterations and weights for stats calculation	MATLAB	(Nielsen 2007)
15	CE	– – – – – -	MATLAB	(Adar et al. 2011; Eismann and Meola 2008; Eismann, Meola, and Hardie 2008; Dekker et al. 2013)
16	CRC	– – – – –	MATLAB	(Eismann and Meola 2008; Eismann, Meola, and Hardie 2008; Adar et al. 2011)

approach proposed by Du, Wasson, and King (2005), first, any pixel vector can be selected as an initial desired object denoted by m_0. A pixel vector that has the largest least square error between itself and m_0 is found and selected as the first object pixel vector denoted by m_1. Then once again, the least square error is applied on the vectors $(\boldsymbol{m_0}\ \boldsymbol{m_1})$, and from the outputs, the pixel vector that yields the largest least square error is selected to be the second object pixel vector m_2. The same procedure with $(\boldsymbol{m_0}\ \boldsymbol{m_1}\ \boldsymbol{m_2})$ is repeated until the resulting least square error is below the prescribed error threshold η. In the second step, the fully least square error method is applied and the abundance map is produced. These two steps are applied to each temporal data set separately. If there are new endmembers, then each pixel containing a significant abundance of these endmembers can be considered as a change pixel. Otherwise, the two corresponding abundance vectors of each pixel are compared, and in order to obtain a change map, a suitable threshold should be selected. The approach (Ertürk and Plaza 2015) starts with estimating the number of endmembers using hyperspectral signal identification by the minimum error (HySime) algorithm and then proceeds by using Simplex Identification via Split Augmented Lagrangian (SISAL) and *N*-finder algorithm endmember extraction on all of the hyperspectral data sets acquired from the same scene at different times. In the next step, the fully constrained least square error is employed to obtain the abundance value for each endmember. Finally, the change map for each endmember is created by a simple difference operation between the corresponding abundance maps incorporating the suitable threshold. The total change map is created by integrating the change maps for each endmember.

2.2.3. *Combining the transformation-based methods with the Anomaly Detector (AD)*

In combining transformation-based methods with the AD algorithm, the first step involves a transformation to maximize the overall correlation between the data sets so that the change areas have higher values and significantly different statistical characteristics that can be identified as anomalies. In the next step, the AD is applied to the transformed image and the extracted changes. In this step, the anomalous differences are detected by comparing the pixels, and thus the change map is created (Pieper et al. 2015a, 2015b). The anomalous differences are detected by comparing the pixels of the two transformed images. Therefore, AD algorithms can be found by the scores in the aggregated image by subtracting each image. After implementing the AD algorithm, changes appear in the foreground with high values, which can be detected by selecting a suitable threshold. The most popular transformation-based methods combined with the AD algorithms are the Canonical Correlation Analyses (CCAs) (Pieper et al. 2015b; Eismann, Meola, and Hardie 2008; Eismann and Meola 2008), MCCA (Pieper et al. 2015b), CE (Pieper et al. 2015b; Eismann and Meola 2008; Eismann, Meola, and Hardie 2008), PCA (Pieper et al. 2015a, 2015b), and SVD (Pieper et al. 2015a, 2015b).

2.3. *Direct classification-based*

The direct classification-based methods (G3) are based on using just one classifier for stacking multi-temporal data sets (Singh 1989; Mas 1999). The *no-change* areas have the same spectral signatures in the multi-temporal data set and the *change* areas have spectral features with significantly different statistical characteristics, which make them detectable by classifiers (Mas 1999). The direct classification-based methods are divided into two main categories: supervised and unsupervised (Table 6). The common methods for the supervised direct classification are Support Vector Machine (SVM), the Minimum Distance (MiD) classifier, the SAM classifier, the Maximum Likelihood Classifier (MLC) (Lee 2011; Mousazadeh et al. 2015; Ghobadi et al. 2015; Sallaba 2009), and supervised Target Change Detection (TCD) (Wu, Zhang, and Bo 2012). The unsupervised methods do not

Table 6. G3 HSCD methods information.

No.	Method name	Setting parameter(s)	Reference software	Reference
1	FCM	Number of iterations and clusters	MATLAB	(Raja et al. 2013; Ahlqvist 2008; Gong, Zhou, and Ma 2012)
2	ISODATA	Number of iterations and clusters is main and other primary parameters	ENVI	(Mas 1999)
3	KM	Number of iterations and clusters is main and other primary parameters	ENVI	(Fan 2008; Fröjse 2011)
4	SAM	Set maximum angle	ENVI	– – – -
5	SVM	Penalty parameter and Gamma in kernel function	ENVI	(Y. Yang and Yan 2015; Sica et al. 2016)
6	MiD	Maximum distance error, maximum standard division from mean	ENVI	– – – –
7	MLE	Set probability threshold and data scale factor	ENVI	(Lee 2011; Mousazadeh et al. 2015; Ghobadi et al. 2015; Sallaba 2009)
8	TCD	– – – – -	MATLAB	(Wu, Zhang, and Du 2012)

Table 7. G4 HSCD methods information.

No.	Method name	Setting parameter(s)	Reference software	Reference
1	FCM	Number of iterations and clusters	MATLAB	(Raja et al. 2013; Ahlqvist 2008; Gong, Zhou, and Ma 2012)
2	ISODATA	Number of iterations and clusters is main and other primary parameters	ENVI	(Omo-Irabor 2016)
3	KM	Number of iterations and clusters is main and other primary parameters	ENVI	(Fan 2008; Fröjse 2011)
4	SAM	Set maximum angle	ENVI	– – – – – –
5	SVM	Penalty parameter and Gamma in kernel function	ENVI	(Y. Yang and Yan 2015; Sica et al. 2016)
6	MiD	Maximum distance error, maximum standard division from mean	ENVI	– – – – – – -
7	MLE	Set probability threshold and data scale factor	ENVI	(Lee 2011; Mousazadeh et al. 2015; Ghobadi et al. 2015; Sallaba 2009)

need training data such as *k*-means (KM), ISODATA (Mas 1999), and fuzzy *c*-means (FCM) (Raja et al. 2013; Ahlqvist 2008; Gong, Zhou, and Jingjing 2012).

2.4. *Post-classification-based*

The post-classification-based (G4) change detection method is a popular approach for change detection analysis (Foody 2002). This group of change detection algorithms uses two or more data sets for creating the change map. The post-classification-based change detection is implemented in two steps: in the first step, each data set is classified individually, and in the second step, the results of each classification are compared with each other (e.g. the ID algorithm used for this purpose), and finally the change map is generated. One of the biggest advantages of this group is that the change map can be created by data sets from different sensors. In other words, because each data set is classified separately, it is not necessary to use a data set from only one sensor. Generally, the post-classification-based methods (Table 7) can be divided into supervised and unsupervised classifiers (Rivera 2005). The difference between these two is that the supervised classifiers need training data for classification (Rivera 2005; Rahman 2016; Rajeswari, Saritha, and Santhosh Kumar 2014). Some popular examples of these methods include MLC (Lee 2011; Mousazadeh et al. 2015; Ghobadi et al. 2015; Sallaba 2009), SVM (Yang and Yan 2016; Sica et al. 2016), the MiD classifier, and the SAM classifier. Conversely, the unsupervised methods do not need a training procedure. Some commonly used examples of these methods include KM (Fan 2008; Fröjse 2011), ISODATA (Omo-Irabor 2016), and FCM (Raja et al. 2013; Ahlqvist 2008; Gong, Zhou, and Jingjing 2012). The main difference between the post-classification-based method and the direct classification-based method is that the former classifies each multi-temporal data set independently, whereas the latter classifies all of the multi-temporal data sets altogether.

2.5. *Hybrid-based*

The hybrid-based methods (G5) are the combination of the previously explained groups and have become increasingly popular among researchers in the field (Naveena and Jij 2015). These methods are applied more to the unsupervised or automatic procedures.

Table 8. G5 HSCD methods information.

No.	Abbreviation	Setting parameter(s)	Reference software	Reference
1	ICA-ID	– – – – – –	MATLAB	(Gulati and Pal 2014)
2	C^2VA	Alpha, number of iterations, and difference to reach the exit EM loop	MATLAB	(Bovolo, Marchesi, and Bruzzone 2012)
3	MSU	Alpha, number of iterations, and difference to reach the exit EM loop and similarity threshold, false rate	MATLAB	(Liu 2015; Liu et al. 2016)
4	SCRC	Number of segments	MATLAB	(Eismann and Meola 2008; Eismann, Meola, and Hardie 2008)
5	SPIM	Parameters for FCM, number of iterations, epsilon for iterations, and initial weights	MATLAB	(Marpu, Gamba, and Benediktsson 2011)
6	SSDM-CD	Two regularization parameters, SVM parameter, and KNN parameters	MATLAB	(Yuan, Lv, and Lu 2015)

An example of the hybrid methods (Table 8) follows these general steps. First, a transformation-based or match-based method is applied, which is followed by detection of *change* and *no-change* areas using a segmentation or clustering method, which results in the generation of the change map. These methods are the ICA-Image Difference (ICA-ID) (Gulati and Pal 2014), compressed change vector analysis (C^2VA) (Bovolo, Marchesi, and Bruzzone 2012), Multi-temporal Spectral Unmixing (MSU) (Liu 2015; Liu et al. 2016), Segment Cross-Covariance (SCRC) (Eismann and Meola 2008; Eismann, Meola, and Hardie 2008), SSDM-CD (Yuan, Lv, and Lu 2015), and the Segment PCA-based IR-MAD (SPIM) (Marpu, Gamba, and Benediktsson 2011).

3. Case study

In this work, we utilized two widely used real-world hyperspectral data sets to evaluate the performance of the HSCD methods. One of the important factors for assessing different types of HSCD methods is the quality of providing the ground truth data. For evaluating different HSCD methods, it is necessary to incorporate a reference data set as the ground truth with high reliability because the false ground truth will lead to incorrect results, especially in the review paper. Therefore, the main reason for choosing these data sets is the availability of related ground truth data. The incorporated data sets have been used in many HSCD studies such as Wu, Bo, and Zhang (2013); Liu et al. (2017); Liu et al. (2016); Liu (2015); Wu, Zhang, and Bo (2012); and Yuan, Lv, and Lu (2015). The ground truth data sets were created by visually analysing the above-mentioned studies. Additionally, by using the Pan image data set from Advanced Land Imager (ALI) sensors that are acquired simultaneously with Hyperion data, the detailed visual comparison binary change map is carried out. The multiple change map is created by using high-resolution Google Earth imagery and spectral change profiles. The first data set (China) belongs to a farmland near the city of Yuncheng Jiangsu province in China, which was acquired on 3 May 2006 and 23 April 2007. This scene is mainly a combination of soil, river, tree, building, road, and agricultural field. The second data set (USA) belongs to an irrigated agricultural field of Hermiston city in Umatilla County, Oregon, OR, the USA, which was acquired on 1 May 2004 and 8 May 2007. The land-cover types are soil, irrigated fields, river, building, type of cultivated land, and grassland.

For both mentioned data sets, all changes in data set related to the type of land cover and river. In addition, both data sets belong to the Hyperion sensor (Figure 3). These two data sets (China and USA) can be found online in http://rslab.ut.ac.ir.

Figure 3 presents both the change map in binary format and the multiple change map. The USA data set has six different change classes and the China data set has three change classes. In two groups, including the supervised direct classification-based methods (G3) and the supervised post-classification-based methods (G4), it is necessary to use training data; therefore, details of the incorporated number of training data for each class are presented in Table 9. The number of training data matters a lot in any statistical estimation technique and machine learning, but there is no standard for a number of training samples. Nevertheless, in this research, we try to keep the number of training data at the sufficient amount. Therefore, all training data for both 'change' and 'no-change' classes selected randomly and distributed spatially cover all of the data sets.

4. Experiments and results

4.1. *Preprocessing*

All HSCD methods implemented in this article are addressed through either ENVI or MATLAB software. See Tables 4 through 8 for more details on the setting parameters for HSCD methods and the related source software. It is worth mentioning that data preprocessing plays a crucial role in the CD process (İlsever and Unsalan 2012). In our work, this step can be divided into two major categories: spectral and spatial corrections. For a better evaluation of the performance of HSCD methods, four scenarios (Table 2) were designed for each data set. The first scenario (S1) is incorporating raw data and just spatial correction without spectral correction. The second scenario (S2) involves utilization of atmospheric and spatial correction without noise reduction. The third scenario (S3) includes merely the noise reduction and spatial correction, and finally, the fourth scenario (S4) comprises a full processing of all above-mentioned preprocessing scenarios. The preprocessing steps start by spectral correction, followed by applying the spatial correction. The first step of preprocessing consists of omitting no-data bands. Through this step, 44 bands (1–7, 58–76, and 225–242) were removed from our imagery (Scheffler and Karrasch 2014). In addition, among the 198 bands, two noisy bands, including 77 and 78, were removed (Shahid et al. 2006; Datt et al. 2003). In the second step, the pixels in sample 129 and all of the lines were shifted to sample 256 in shortwave infrared spectral bands (Goodenough et al. 2003; Li et al. 2008). The third step is de-striping and removing the zero-line step by utilizing the means and the global approach. The fourth step of preprocessing includes the de-noising process, which, in this study, is executed by the PCA wavelet (Lam, Sato, and Sato 2012) method. The fifth step of the preprocessing procedure is the de-smiling process, which aims to remove the effect of the application, and is executed using the ENVI cross-track illumination correction module. The sixth step of the preprocessing procedure is radiometric correction. To achieve this objective, the digital number of the pixels is converted into physical radiance. This step of preprocessing can be carried out using Equation 1 for band numbers 1–70 and Equation 2 for the remaining band numbers 71–242.

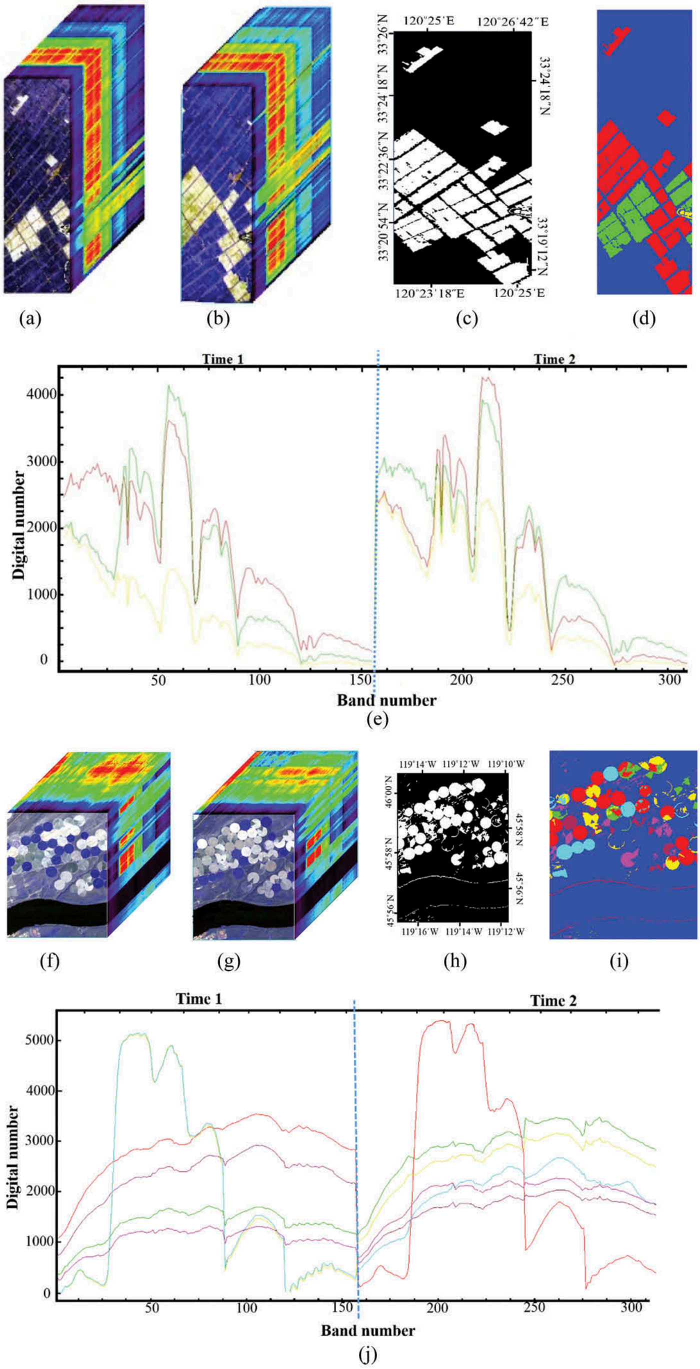


Figure 3. (a) and (b) present false-colour composites of the original hyperspectral images acquired in 2006 and 2007 of the China data set, respectively; (c) binary change map ground truth; (d) multiple change map ground truth; and (e) the spectral change profile for the China data set. (f) and (g) False-colour composite of the original hyperspectral images acquired in 2004 and 2007 of the USA data set, respectively; (h) binary change map ground truth; (i) multiple change map ground truth; and (g) the spectral change profile for the USA data set.

Table 9. Legend and training data for supervised classification methods in G3 and G4.

Data set	Class	Symbol	Label colour	Number of pixels	Number of training data
USA	*Change*	ω_1		5896	608
		ω_2		1524	612
		ω_3		2905	427
		ω_4		2744	732
		ω_5		1808	515
		ω_6		1850	419
	No-Change	C		57,501	3919
China	*Change*	ω_1		12,229	2557
		ω_2		6019	1634
		ω_3		145	41
	No-Change	C		40,407	4875

$$\mathrm{Radiance_i} = (\mathrm{DN})_i/40 \qquad i = 1:70. \tag{1}$$

$$\mathrm{Radiance_i} = (\mathrm{DN})_i/80 \qquad i = 71:242. \tag{2}$$

The next step of preprocessing is atmospheric correction, which is an extremely critical process. For this purpose, we used the FLAASH model of ENVI software, and then applied band selection. Therefore, 154 bands were selected in total as the input data sets to compare the HSCD methods. The final step of preprocessing of the hyperspectral data set is a spatial correction that was applied based on relative correction (image to image). For spatial or geometric correction, the ENVI image registration tool was used, and then, several common control points were selected and a second-order polynomial was utilized for resampling purposes. The resulting accuracy of the geometric correction (RMSE) is about 0.4 pixel.

Steps one, two, and three and the spatial correction are common preprocessing steps that are applied in all four scenarios. We used three indicators to assess the HSCD algorithms in transformation-based, match-based, and some of the hybrid-based methods: the overall accuracy, kappa coefficient (κ), and the Area Under the Curve (AUC). Therefore, in the first step, we found the minimum and maximum outputs of the initial change map and then selected the threshold in the interval within 0.01 × (maximum – minimum). In the next step, we calculated the corresponding accuracy based on the selected threshold on ground truth. Among the different calculated accuracies, both maximum accuracy and the corresponding threshold were selected based on the comparison of each change map with the incorporated threshold on ground truth. Therefore, in this evaluation, ground truth is necessary and not feasible in real applications due to the unavailability of ground truth data. Therefore, the main purpose of this work is only to assess the potential and performance of the utilized methods for detecting changes. In other words, this assessment reveals the maximum accuracy that each method could reach. Owing to the unavailability of ground truth for the change detection problem, the unsupervised threshold selection methods must be considered. The HSCD methods based on structure aspect have two output types: (1) single band and (2) cube data. Therefore, in this study, the Otsu thresholds selection algorithm (Otsu 1979) is used for single band structures and the KM algorithm is used for the cube data.

4.2. *Experimental results*

In this part, the experimental results of the HSCD methods in two real hyperspectral data sets are evaluated, analysed, and compared for the five HSCD groups and the four scenarios. The numerical analysis of the results of match-based methods (G1), transformation-based methods (G2), and hybrid-based methods (G5) is on the binary map. In addition, the results of post-classification (G4) and direct classification-based (G3) methods are on the multiple change map.

4.2.1. *Result of the match-based group (G1)*

The result of the match-based methods considered in two scenarios is explained in the next sections.

4.2.1.1. ***Supervised threshold selection outputs.*** The results of match-based methods presented in Table 10 and Figures 4 and 5 show the numerical and visual analyses of the performance of similarity-based methods for both USA and China data sets in different scenarios and groups. The change detection result of the match-based methods is evaluated in the visual and numerical analyses. Figures 4 and 5 show the best results for each method in group (G1) and in different scenarios with supervised threshold selection. Table 10 presents the results of the match-based change detection methods. It is clear that the match-based methods have a good efficiency in the main change areas in both data sets. The difference in performance among the methods relates to subtle changes. The output of similarity-based methods is in the range of (0, 1), whereas the output of other match-based methods do not have a restriction and can cover a wide range of values. The results of ID and IR methods have a number of bands equal to the input data, whereas the other methods in G1 group have only single bands. The best results for the China data set located in S1 for most methods, whereas the best result for the USA data set located in other scenarios (S2 and S3). These variations are based on preprocessing and these issues refer to the important effect of preprocessing in the HSCD. The most informative bands in ID and IR methods are different and are not the same for all algorithms. Therefore, the informative bands could be different depending on the virtual structure of the algorithm based on 'band–band' formation. For example, in the China data set, for the ID and IR methods in S4, the number of informative bands is 121 and 5, respectively. Therefore, the informative bands of the ID and IR algorithms located in principles 118 and 4 for the China data set (S2), respectively. Therefore, the informative band is different for each algorithm that has a structure based on band–band formation. In addition, this proposition is clearer for ID algorithm among scenarios, whereas the issue is tangible in the IR method.

The best performances of each method based on visual and numerical analyses in the China data set are SAM (S1), ID (S1), IR (S1), CCO (S1), MD (S4), and ED (S1), and for the USA data set they are SAM (S3), ID (S2), IR (S3), CCO (S2), MD (S4), and ED (S2). The SAM and MD algorithms have a high tolerance with respect to the type of scene, as these methods highlighted the main changes in the China data set, whereas there are variations on detection of changes in the USA data set. In addition, as it is clear especially on the USA data set, these methods have low performance compared to other match-based methods. This issue is more dominant in river class type that is highlighted by these

Table 10. Result performance of G1 for both data sets in all scenarios.

			Overall accuracy (%)		κ		
Method	Scenario	Data set	Supervised	Unsupervised	Supervised	Unsupervised	AUC
SAM	# 1	USA	77.53	69.57	0.175	0.219	0.556
		China	97.83	89.67	0.949	0.736	0.995
	# 2	USA	79.16	75.56	0.377	0.356	0.754
		China	91.76	90.49	0.808	0.785	0.953
	# 3	USA	90.21	82.47	0.691	0.576	0.920
		China	93.32	89.97	0.843	0.748	0.960
	# 4	USA	87.09	80.27	0.560	0.458	0.755
		China	92.85	89.97	0.835	0.748	0.966
ID	# 1	USA	92.92	89.05	0.795	0.677	0.949
		China	96.36	95.12	0.915	0.883	0.991
	# 2	USA	96.24	93.12	0.891	0.816	0.982
		China	92.62	90.13	0.808	0.766	0.963
	# 3	USA	95.64	92.97	0.876	0.800	0.980
		China	92.91	92.61	0.837	0.825	0.966
	# 4	USA	95.14	91.15	0.858	0.710	0.965
		China	92.75	92.61	0.835	0.825	0.966
IR	# 1	USA	87.43	86.18	0.573	0.558	0.722
		China	95.88	96.33	0.905	0.915	0.647
	# 2	USA	91.74	91.69	0.748	0.730	0.906
		China	91.35	90.69	0.801	0.789	0.954
	# 3	USA	92.65	91.51	0.787	0.736	0.949
		China	91.72	91.76	0.811	0.812	0.959
	# 4	USA	91.18	84.62	0.719	0.580	0.849
		China	91.28	91.76	0.802	0.812	0.956
CCO	# 1	USA	86.87	85.19	0632	0.533	0.904
		China	97.31	97.03	0.938	0.930	0.994
	# 2	USA	98.47	93.52	0.957	0.830	0.997
		China	90.22	90.15	0.775	0.774	0.946
	# 3	USA	94.12	92.91	0.831	0.783	0.967
		China	91.98	92.00	0.815	0.815	0.958
	# 4	USA	96.05	95.61	0.886	0.874	0.980
		China	92.32	92.00	0.825	0.815	0.963
MD	# 1	USA	80.04	79.46	0.632	0.144	0.464
		China	68.77	65.87	0.298	0.298	0.670
	# 2	USA	77.83	79.84	0.347	0.248	0.600
		China	68.58	62.32	0.223	0.103	0.618
	# 3	USA	79.46	64.49	0.413	0.230	0.656
		China	68.58	62.76	0.240	0.123	0.623
	# 4	USA	84.86	68.14	0.535	0.284	0.685
		China	75.32	62.76	0.476	0.123	0.237
ED	# 1	USA	87.67	87.25	0.642	0.640	0.898
		China	97.07	94.30	0.923	0.861	0.993
	# 2	USA	98.19	95.86	0.949	0.875	0.997
		China	89.97	92.26	0.770	0.820	0.941
	# 3	USA	94.57	90.46	0.845	0.756	0.982
		China	91.84	91.85	0.811	0.810	0.955
	# 4	USA	95.37	89.05	0.866	0.851	0.977
		China	92.35	91.85	0.825	0.810	0.963

methods. The other algorithms are less sensitive to preprocessing and type of scene and have the same trend. The ID and IR methods have different outputs compared to other methods, and this difference relates to the structure of these algorithms. One of the most important reasons for the good efficiency of CC and ED methods is the ability of these methods to separate the *change* from *no-change* area compared to the other methods. In other words, lack of good prediction caused mixing of the *change* pixels with the *no-change* ones, thereby having a negative effect on change detection result.

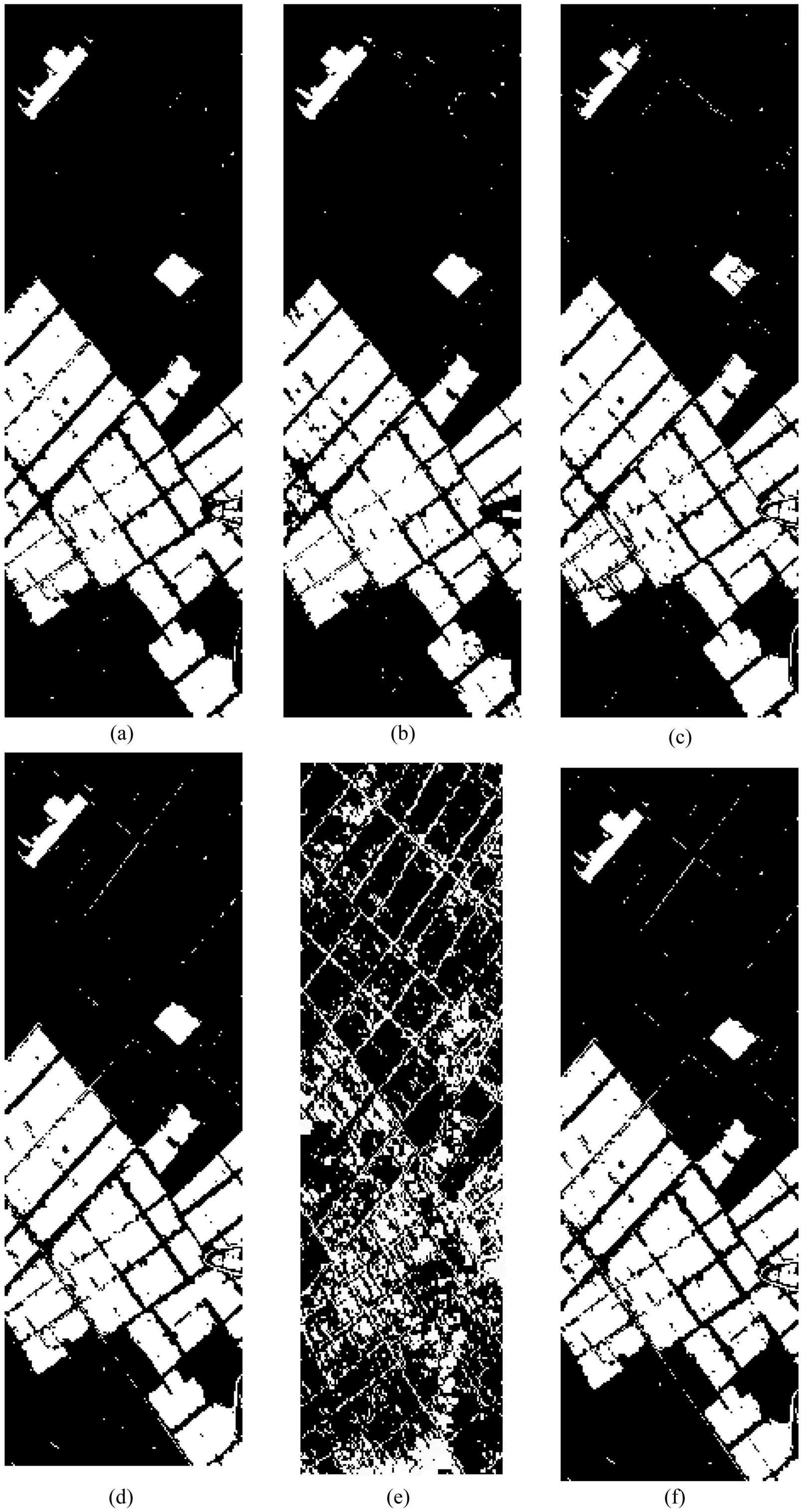


Figure 4. Result of the best performance in G1 for all scenarios in the China data set using supervised threshold selection: (a) SAM; (b) ID; (c) IR; (d) CCO; (e) MD; and (f) ED.

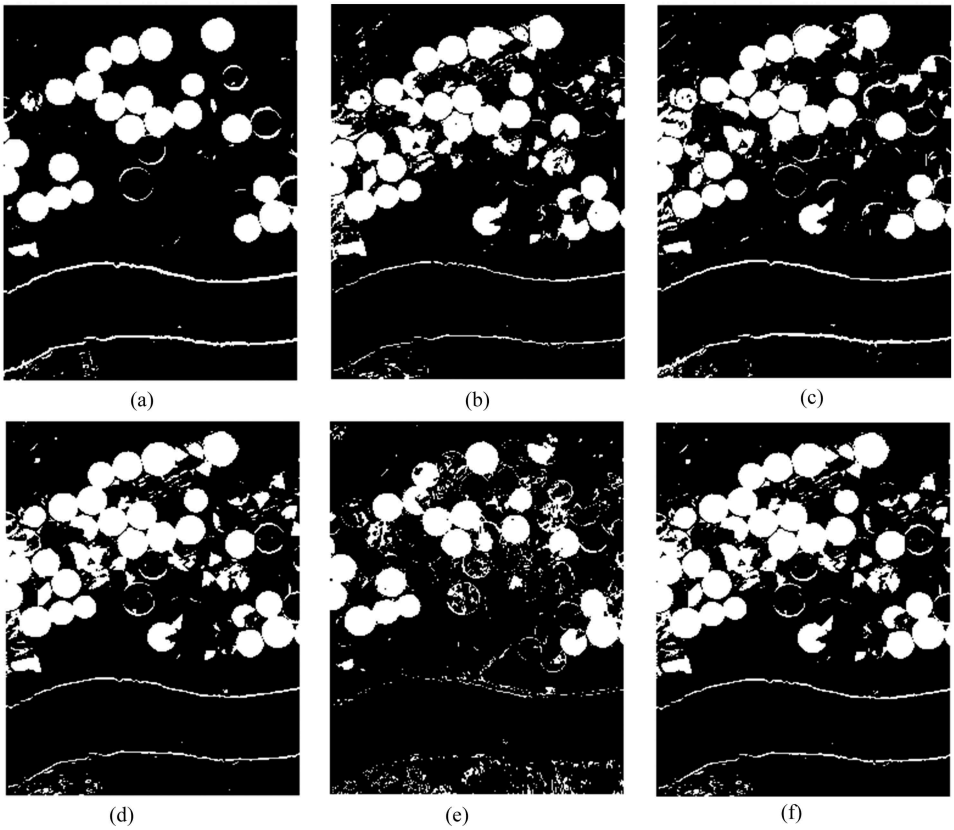


Figure 5. Result of the best performance in G1 for all scenarios in the USA data set using supervised threshold selection: (a) SAM; (b) ID; (c) IR; (d) CCO; (e) MD; and (f) ED.

4.2.1.2. ***Unsupervised threshold selection outputs.*** The result of match-based methods is presented in Table 10 for both USA and China in the unsupervised framework. The presented results in Figures 6 and 7 are provided based on the detection of multiple changes. The main difference between the utilized methods is due to the ability of separation of the *change* area from the *no-change* area. In addition, some methods have a good prediction on the type of change class. In this group, the IR and ID methods have output cube data, so KM clustering is used for these methods. The SAM, MD, ED, and CCO have an output single band and therefore Otsu is used as the threshold algorithm.

The best performance of each method based on visual and numerical analyses in the China data set are SAM (S2), ID (S1), IR (S1), CCO (S1), MD (S1), and ED (S1), and for the USA data set they are SAM (S3), ID (S2), IR (S2), CCO (S4), MD (S1), and ED (S2). The main difference among the types of match-based methods is focusing on the multiple changes and thresholding (the difference between supervised and unsupervised threshold selection). Almost all match-based methods have different results among supervised and unsupervised threshold selection, and this theme refers to the role of threshold selection on the change detection results. These methods have lower accuracy in unsupervised threshold selection compared to supervised thresholding framework.

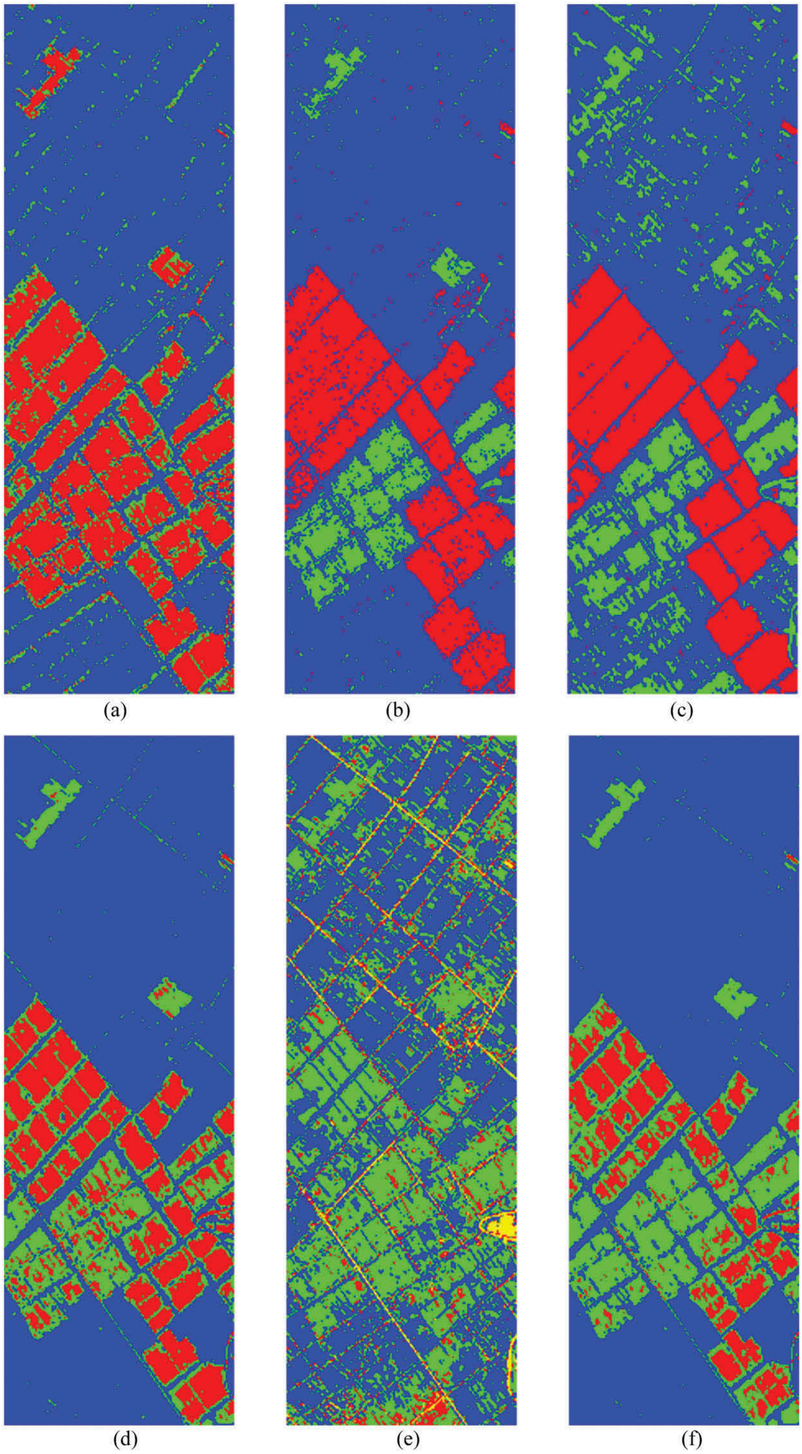

(a) (b) (c)
(d) (e) (f)

Figure 6. Result of the best performance in G1 for all scenarios in the China data set using unsupervised threshold selection: (a) SAM; (b) ID; (c) IR; (d) CCO; (e) MD; and (f) ED.

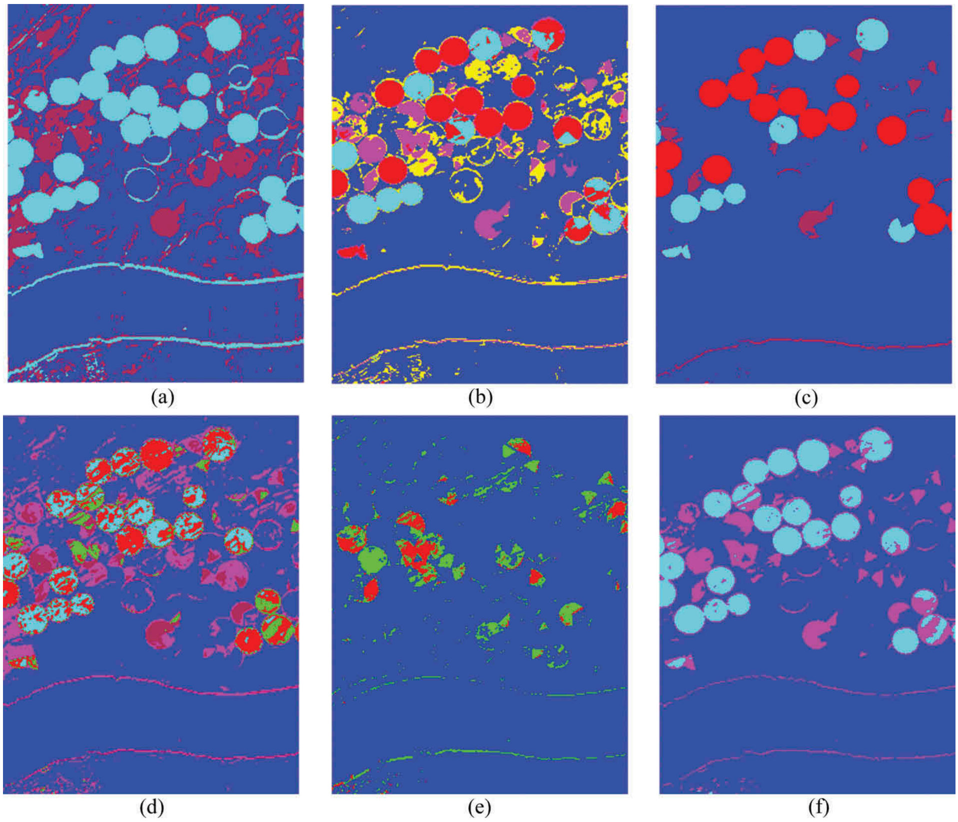


Figure 7. Result of the best performance in G1 for all scenarios in the USA data set using unsupervised threshold selection: (a) SAM; (b) ID; (c) IR; (d) CCO; (e) MD; and (f) ED.

This is clear for some methods in utilized scenarios, e.g. SAM (S1) and MiD (S4) for the China data set and SAM (S4), IR(S4), CCO (S2), MD (S4), and ED (S3 and S4) for the USA data set. However, there are special cases where unsupervised thresholding has a better result (binary change map) compared to supervised thresholding. This issue is evident in the IR algorithm (S1 and S4) and ED algorithms (S2) for the China data set and in the MiD algorithm (S2) for the USA data set.

Since we have a certain number of classes in the ground truth map, it is expected that the output of the algorithm is inconsistent with the ground truth map. Therefore, this evaluation is focused more on two main issues: (1) accuracy of change detection of algorithms for unsupervised thresholding and (2) potentiality of algorithms in the detection of change types. Instead of extracting multiple changes, these methods are focused on labelling the background (*no-change* class) in different clusters. The multiple changes detected by unsupervised thresholding for the China data set are SAM (ω_1, ω_2), ID (ω_1, ω_2), IR (ω_1, ω_2), CCO (ω_1, ω_2), MD (ω_1, ω_2, ω_3), and ED (ω_1, ω_2). Moreover, for the USA data set they are SAM (ω_4, ω_6), ID (ω_3, ω_4, ω_5, ω_6), IR (ω_1, ω_4, ω_6), CCO (ω_1, ω_2, ω_3, ω_4, ω_5, ω_6), MD (ω_1, ω_2), and ED (ω_1, ω_6). In the China data set, all multiple changes are detected by unsupervised multiple thresholding, whereas in the USA data set, only the two methods of SAM and CCO could detect multiple changes completely.

4.2.2. *Results of the transformation-based group (G2)*

All transformation-based methods are considered in two frameworks as before: (1) supervised threshold selection and (2) unsupervised threshold selection. The next section will explain the experimental results in more detail.

4.2.2.1. ***Supervised threshold selection outputs.*** The experimental results for transformation-based methods are presented in Table 11 and Figures 8 and 9 for both USA and China data sets in different scenarios and groups by considering supervised threshold selection. The general transformation-based methods are investigated in the visual and numerical analyses. The CEN and CEC have weak results and different performances in different types of land covers and scenarios. The best achieved performances based on visual and numerical analyses in the China data set are CEN (S1), CEC (S1), ICA (S1), CCA-AD (S1), CE-AD (S4), MCCA-AD (S1), PCA-AD (S1), SVD-AD (S1), *N*-finder (S1), SISAL (S1), LSQ (S2), PCA (S1), MAD (S1), IR-MAD (S1), CE (S1), CRC (S1), SSB (S1), and MLE (S4). For the USA data set, the best performances are achieved by CEN (S1), CEC (S1), ICA (S2), CCA-AD (S3), CE-AD (S4), MCCA-AD (S1), PCA-AD (S3), SVD-AD (S3), *N*-finder (S4), SISAL (S1), LSQ (S2), PCA (S2), MAD (S1), IR-MAD (S2), CE (S1), CRC (S2), SSB (S3), and MLE (S3).

The SSB, MAD, and ICA methods are sensitive to waterbody class type; this class theme is evident on the river in the USA data set. The SSB and IR-MAD methods signalled the *change* area from the *no-change* one; in other words, the foreground (*change* area) simply separated from the background. The PCA, CRC, and CE methods have similar trends in detecting changes in the data set by considering the same effect on preprocessing. The CRC, CE, MAD, and IR-MAD have the same structure ('band-band') in this group. These methods, compared to PCA and ICA, do not consider the criterion for finding informative bands. The number of informative bands of the China data set in S2 for MAD, IR-MAD, CRC, and CE is 61, 146, 121, and 5, respectively. In addition, the number of informative bands in this data set in S4 for MAD, IR-MAD, CE, and CRC is 59, 147, 25, and 118, respectively.

It is clear that the change detection methods based on AD predictors are extremely sensitive to preprocessing. Because all of the spectral bands put as input for these methods, these methods do not give a good result. Therefore, for using these methods, utilizing dimension reduction procedures and removing systematic errors are necessary. The SISAL and *N*-finder based on unmixing change detection methods have different results in two data sets, whereas both of these methods used the same endmember estimation and employed one algorithm to obtain the abundance value for each endmember. This issue shows the importance of endmember extraction methods in HSCD. In addition, these methods give a different response in different scenarios, and these differences are due to the utilized methods for estimating and extracting the endmember and estimating the abundance vector. Moreover, in unmixing algorithms based on optimization procedures such as *N*-finder endmember extraction, different change detection results are generated in each execution. The issue refers to the role of endmember extraction in the unmixing change detection. The LSQ method has a less similar trend compared to SISAL and *N*-finder. The IR-MAD algorithm has low variation in atmospheric condition. The main reason is due to incorporating algorithms, which are invariant to linear and affine transformations of the original data.

4.2.2.2. *Unsupervised threshold selection outputs.*
The result of transformation-based methods in unsupervised threshold selection is presented in Table 11 and Figures 10 and 11 for both USA and China data sets in different scenarios and groups. Similar to the match-based methods, the thresholding theme of all transformation-based methods is based on single band and cube clustering. The transformation-based method's output has a single band as follows: CEN, CEC, CCA-AD, CE-AD, MCCA-AD, PCA-AD, SVD-AD, *N*-finder, SISAL, LSQ, CRC, SSB, and MLE, whereas the other transformation-based methods have output cube data.

Based on the visual and numerical analyses, the best achieved performances in the China data set for unsupervised framework were CEN (S1), CEC (S1), ICA (S1), CCA-AD (S3), CE-AD (S1), MCCA-AD (S1), PCA-AD (S2), SVD-AD (S2), *N*-finder (S1), SISAL (S1), LSQ (S2), PCA (S1), MAD (S1), IR-MAD (S1), CE (S1), CRC (S4), SSB (S1), and MLE (S4). For the USA data set, the best performance was achieved by CEN (S2), CEC(S1), ICA(S3), CCA-AD (S2), CE-AD (S2), MCCA-AD (S4), PCA-AD (S2), SVD-AD (S4), *N*-finder (S4), SISAL (S4), LSQ (S1), PCA (S4), MAD (S3), IR-MAD (S2), CE (S2), CRC (S2), SSB (S3), and MLE (S2). In some methods, the best performances are the same for the scenarios in two frameworks of supervised and unsupervised thresholding. However, this issue is different for most methods. These algorithms for the China data set are CCA-AD, MCCA-AD, and PCA-AD. Furthermore, for the USA data set, the algorithms are SISAL, LSQ, PCA, MAD, IR-MAD, CE, and MLE. There are some variations between the two threshold selection techniques (supervised and unsupervised) for most transformation-based methods in different scenarios. However, this issue is tangible for SSB and MLE.

Similar to match-based methods, the transformation-based methods have different performances in predicting multiple changes. The results of multiple changes for each of the methods are different. The numbers of change class detected by multiple thresholding methods for each method in the China data set are listed as follows: CEN (ω_1, ω_2), CEC (ω_1, ω_2), ICA (ω_1, $\omega\omega_2$), CCA-AD (ω_1, $\omega\omega_2$), CE-AD (ω_1, ω_2), MCCA-AD (ω_1, ω_2), PCA-AD (ω_1, ω_2), SVD-AD (ω_1, ω_2), *N*-finder (ω_1, ω_2, ω_3), SISAL (ω_1, $\omega\omega_2$), LSQ (ω_2), PCA (ω_1, ω_2, ω_3), MAD (ω_1, ω_2), IR-MAD (ω_1, ω_2, ω_3), CE (ω_1, ω_2), CRC (ω_1, ω_2, ω_3), SSB (ω_1, ω_2, ω_3), and MLE (ω_1, ω_2, ω_3). Besides, for the USA data set, the numbers are as follows: CEN (ω_4), CEC (ω_4), ICA (ω_3, ω_4), CCA-AD (ω_1, ω_4), CE-AD (ω_1, ω_4), MCCA-AD (ω_3, ω_4), PCA-AD (ω_3, ω_4), SVD-AD (ω_3, ω_4), *N*-finder (ω_1, ω_4, ω_6), SISAL (ω_1, ω_4), LSQ (ω_3, ω_4, ω_6), PCA (ω_1, ω_2, ω_4, ω_6), MAD (ω_1, ω_3, ω_4, $\omega_{5,}$ ω_6), IR-MAD (ω_1, ω_2, ω_3, ω_4, ω_5), CE (ω_3, ω_4, ω_5), CRC (ω_3), SSB (ω_1, ω_4), and MLE (ω_1, ω_4). The visual analysis results for multiple change class showed good performance for most methods in the China data set, whereas in the USA data set, most methods focused on clustering *no-change* class. The PCA and LSQ methods have good performance in detecting multiple changes. The maximum accuracy on the China data set in all groups related to the SSB method and for the USA data set it related to the CRC method.

4.2.3. *Results of the direct classification-based group (G3)*

The experimental results of the direct classification-based methods are presented in Table 12 and Figures 12 and 13 for both USA and China data sets in all scenarios.

Table 11. Result performance of G2 for both data sets in all scenarios.

			Overall accuracy (%)		κ		
Method	Scenario	Data set	Supervised	Unsupervised	Supervised	Unsupervised	AUC
CEN	S1	USA	77.27	76.49	0.272	0.252	0.594
		China	86.19	76.68	0.665	0.432	0.837
	S2	USA	73.70	68.45	0.161	0.171	0.552
		China	79.76	66.22	0.548	0.231	0.847
	S3	USA	75.64	74.69	0.186	0.269	0.547
		China	83.33	74.27	0.597	0.306	0.435
	S4	USA	75.97	73.25	0.202	0.240	0.566
		China	80.23	72.90	0.532	0.403	0.830
CEC	S1	USA	77.27	76.49	0.279	0.260	0.584
		China	86.90	75.69	0.690	0.421	0.848
	S2	USA	74.02	66.33	0.160	0.135	0.540
		China	80.23	65.37	0.544	0.207	0.796
	S3	USA	75.64	74.73	0.186	0.230	0.546
		China	83.80	74.57	0.607	0.312	0.834
	S4	USA	75.64	70.79	0.204	0.191	0.568
		China	81.66	72.92	0.545	0.452	0.831
ICA	S1	USA	89.10	84.01	0.641	0.585	0.813
		China	96.03	95.93	0.908	0.620	0.831
	S2	USA	89.36	85.56	0.688	0.627	0.921
		China	84.57	92.84	0.610	0.835	0.785
	S3	USA	86.94	88.00	0.550	0.590	0.775
		China	85.51	94.15	0.644	0.866	0.844
	S4	USA	88.77	85.86	0.669	0.632	0.918
		China	83.42	92.01	0.597	0.813	0.821
CCA-AD	S1	USA	85.47	85.20	0.566	0.500	0.810
		China	90.48	84.52	0.786	0.680	0.948
	S2	USA	92.27	91.81	0.774	0.760	0.950
		China	89.72	87.17	0.770	0.726	0.937
	S3	USA	92.51	91.11	0.784	0.719	0.950
		China	89.93	89.92	0.775	0.774	0.937
	S4	USA	92.49	91.36	0.776	0.726	0.938
		China	88.57	81.92	0.729	0.534	0.919
CE-AD	S1	USA	86.92	83.84	0.581	0.434	0.748
		China	87.98	93.36	0.836	0.847	0.919
	S2	USA	92.13	92.17	0.772	0.772	0.948
		China	86.90	87.38	0.6749	0.7233	0.837
	S3	USA	81.25	83.84	0.545	0.385	0.861
		China	88.80	88.83	0.725	0.751	0.853
	S4	USA	92.32	90.27	0.772	0.684	0.935
		China	89.61	88.90	0.768	0.762	0.936
	S1	USA	90.22	82.00	0.691	0.422	0.896
		China	96.00	94.87	0.906	0.876	0.978
MCCA-AD	S2	USA	87.71	84.35	0.649	0.480	0.929
		China	89.94	88.24	0.763	0.7354	0.926
	S3	USA	86.84	84.86	0.638	0.480	0.918
		China	91.56	90.56	0.802	0.785	0.939
	S4	USA	87.62	87.01	0.672	0.621	0.924
		China	91.51	90.55	0.800	0.771	0.935
PPCA-AD	S1	USA	91.33	85.13	0.757	0.520	0.930
		China	96.23	93.40	0.912	0.840	0.981
	S2	USA	91.76	91.76	0.756	0.748	0.936
		China	90.61	88.52	0.779	0.744	0.933
	S3	USA	92.43	87.05	0.782	0.640	0.949
		China	91.96	90.79	0.812	0.790	0.943
	S4	USA	90.06	86.79	0.699	0.640	0.887
		China	92.12	91.01	0.815	0.783	0.943

(*Continued*)

Table 11. (Continued).

Method	Scenario	Data set	Overall accuracy (%)		κ		AUC
			Supervised	Unsupervised	Supervised	Unsupervised	
SVD-AD	S1	USA	88.41	83.13	0.640	0.466	0.885
		China	96.17	90.20	0.910	0.786	0.980
	S2	USA	89.42	89.46	0.704	0.701	0.932
		China	90.39	90.55	0.774	0.788	0.932
	S3	USA	90.79	84.18	0.715	0.402	0.856
		China	91.87	85.64	0.809	0.645	0.942
	S4	USA	89.93	90.60	0.709	0.720	0.922
		China	91.80	88.07	0.808	0.723	0.938
SISAL	S1	USA	85.99	85.03	0.526	0.529	0.758
		China	93.56	92.31	0.849	0.816	0.965
	S2	USA	79.85	72.98	0.444	0.378	0.739
		China	79.39	73.82	0.523	0.473	0.854
	S3	USA	-	-	-	-	-
		China	-	-	-	-	-
	S4	USA	77.39	87.57	0.013	0.572	0.512
		China	92.91	87.15	0.837	0.727	0.929
N-finder	S1	USA	84.72	81.11	0.498	0.841	0.852
		China	95.55	91.52	0.896	0.808	0.979
	S2	USA	89.61	89.33	0.444	0.676	0.926
		China	82.24	73.62	0.690	0.481	0.863
	S3	USA	-	-	-	-	-
		China	-	-	-	-	-
	S4	USA	90.80	87.22	0.590	0.656	0.939
		China	93.740	86.21	0.853	0.707	0.937
LSQ	S1	USA	88.88	82.49	0.657	0.588	0.946
		China	78.01	77.57	0.494	0.500	0.825
	S2	USA	89.35	80.73	0.674	0.549	0.931
		China	92.47	92.33	0.828	0.827	0.954
	S3	USA	77.47	73.50	0.392	0.372	0.819
		China	68.84	68.71	0.009	0.009	0.502
	S4	USA	77.39	69.69	0.126	0.188	0.819
		China	92.48	92.33	0.828	0.827	0.924
PCA	S1	USA	94.53	89.34	0.787	0.687	0.941
		China	94.91	86.31	0.882	0.701	0.977
	S2	USA	96.16	89.66	0.887	0.647	0.980
		China	90.49	83.47	0.781	0.645	0.941
	S3	USA	95.79	92.97	0.879	0.805	0.981
		China	92.93	84.92	0.836	0.676	0.958
	S4	USA	95.94	93.48	0.882	0.824	0.969
		China	83.09	83.71	0.575	0.645	0.790
MAD	S1	USA	86.09	82.98	0.574	0.691	0.906
		China	96.10	95.07	0.910	0.887	0.984
	S2	USA	88.32	80.25	0.634	0.512	0.923
		China	92.48	90.90	0.828	0.787	0.958
	S3	USA	88.61	87.18	0.624	0.586	0.927
		China	93.33	86.25	0.848	0.701	0.964
	S4	USA	88.57	82.33	0.618	0.506	0.910
		China	92.99	81.71	0.840	0.582	0.959
IR-MAD	S1	USA	90.94	87.50	0.694	0.603	0.901
		China	96.11	87.48	0.910	0.735	0.984
	S2	USA	92.62	90.57	0.769	0.685	0.940
		China	92.62	85.69	0.820	0.700	0.948
	S3	USA	91.70	89.10	0.752	0.651	0.941
		China	90.40	80.33	80.23	0.438	0.902
	S4	USA	90.87	87.18	0.732	0.573	0.943
		China	92.38	83.28	0.808	0.579	0.926

(*Continued*)

Table 11. (Continued).

Method	Scenario	Data set	Overall accuracy (%)		κ		AUC
			Supervised	Unsupervised	Supervised	Unsupervised	
CE	S1	USA	95.98	85.51	0.819	0.655	0.958
		China	97.09	95.74	0.971	0.898	0.970
	S2	USA	95.95	93.72	0.881	0.828	0.979
		China	93.23	89.26	0.844	0.766	0.969
	S3	USA	95.61	93.13	0.874	0.791	0.830
		China	94.36	90.68	0.869	0.794	0.978
	S4	USA	95.84	93.26	0.878	0.655	0.9689
		China	94.39	91.65	0.870	0.81	0.943
CRC	S1	USA	93.62	91.98	0.770	0.753	0.945
		China	96.26	84.28	0.913	0.605	0.989
	S2	USA	96.06	96.05	0.885	0.981	0.979
		China	92.47	86.77	0.825	0.699	0.954
	S3	USA	95.79	90.78	0.879	0.715	0.979
		China	92.75	83.50	0.832	0.575	0.960
	S4	USA	95.98	92.29	0.882	0.798	0.969
		China	92.84	91.30	0.834	0.802	0.928
SSB	S1	USA	77.50	74.68	0.320	0.188	0.450
		China	97.82	97.14	0.188	0.941	0.995
	S2	USA	79.12	77.92	0.04	0.335	0.752
		China	91.82	90.11	0.408	0.779	0.953
	S3	USA	90.26	89.99	0.006	0.673	0.918
		China	93.35	92.26	0.843	0.811	0.959
	S4	USA	87.12	86.77	0.599	0.535	0.752
		China	92.90	92.63	0.835	0.832	0.929
MLE	S1	USA	77.56	78.13	0.346	0.476	0.204
		China	78.173	78.19	0.439	0.453	0.780
	S2	USA	83.69	82.65	0.125	0.489	0.846
		China	77.06	76.26	0.808	0.409	0.746
	S3	USA	85.21	82.39	0.525	0.514	0.867
		China	78.12	76.53	0.403	0.422	0.757
	S4	USA	84.195	82.04	0.123	0.488	0.846
		China	92.16	91.06	0.812	0.784	0.921

Generally, the direct classification-based methods are investigated in the visual and numerical analyses. The direct classification-based methods are applied in two categories – supervised and unsupervised. The main concern in the supervised methods is providing suitable training data. For more details on the training data, see Table 2. Therefore, all *no-change* classes give the same label and other *change* classes give different labels. The main reason for using large training data is to minimize the effect of the Hughes phenomenon so there is no need to apply dimensional reduction. Some methods such as MLC need to estimate the covariance matrix and this caused the number of training data to be more than the dimensional data set. Thus, some of the classes have a lower number of training data sets than the dimensional data, which are deleted. The setting parameters for the supervised methods are set via cross-validation. The SVM algorithm has two parameters (γ, and C). The RBF kernel parameter is (γ = 0.008, 0.21) and the penalty coefficient is (C = 210, 280) for the China and USA data sets. The 'set max angle' is the parameter (threshold) of the SAM algorithm. If the calculated SAM value for each pixel is greater than the threshold, then the pixel values cannot be classified. For this purpose, the best threshold for all classes in two data sets is set to 0.45 for all of the scenarios. The MLC classifier has two setting parameters; the probability threshold is set to 0.05 for all classes and the data scale factor is set to the maximum value of each data set. The maximum distance error in the MiD method is set to 10^6 and the maximum standard division from

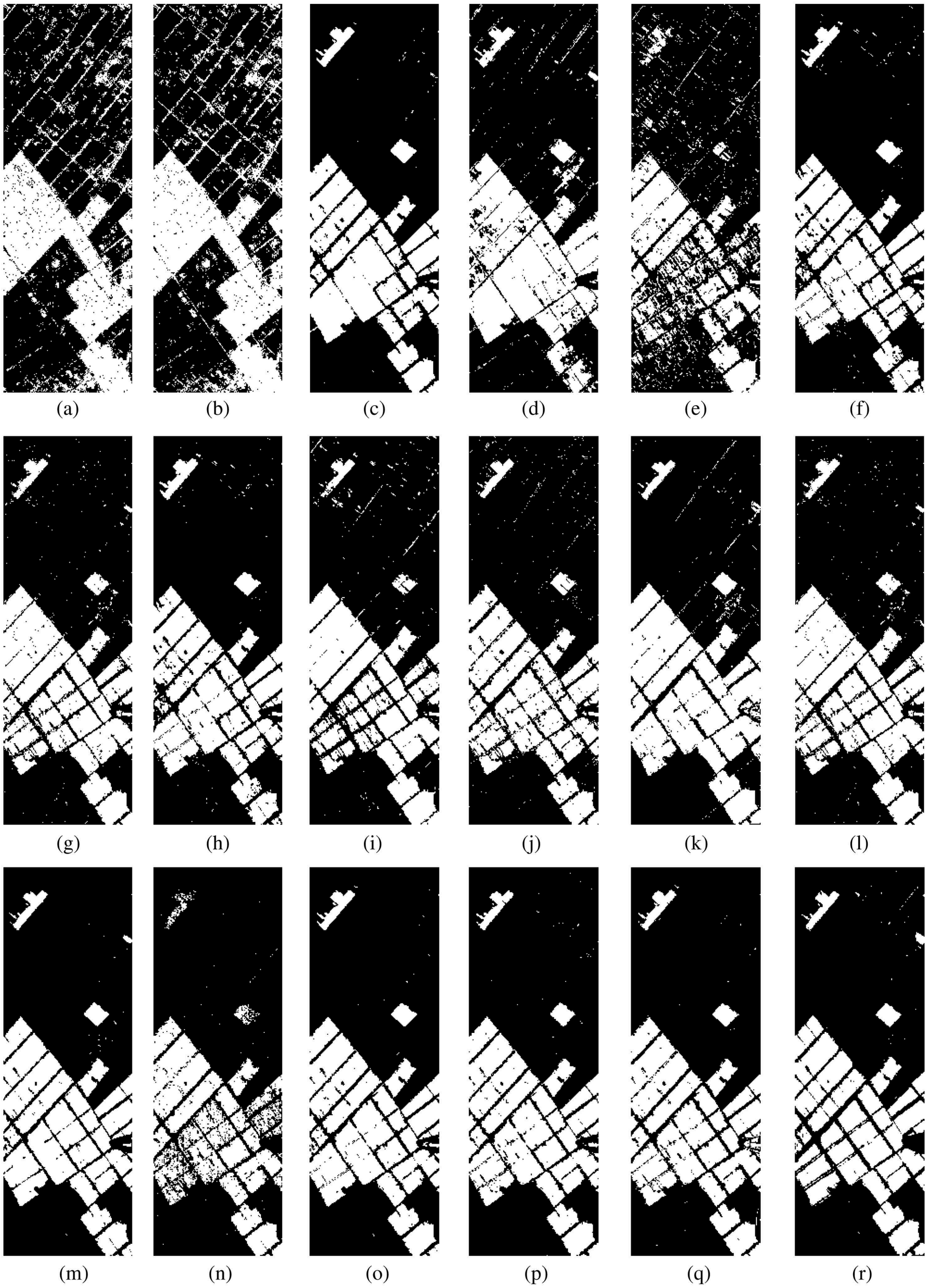


Figure 8. Result of the best performance in G2 for all scenarios in the China data set using supervised threshold selection: (a) CEN; (b) CEC; (c) ICA; (d) CCA-AD; (e) CE-AD; (f) MCCA-AD; (g) PCA -AD; (h) SVD-AD; (i) SISAL; (j) *N*-finder; (k) LSQ; (l) PCA; (m) MAD; (n) IR-MAD; (o) CE; (p) CRC; (q) SSB; and (r) MLE.

the mean is set to 10^3. The TCD methods have no setting parameter but the number of components for dimensional reduction in the China and USA data sets is set to 8 and 12, respectively. For the TCD method estimation, a suitable threshold for detecting target

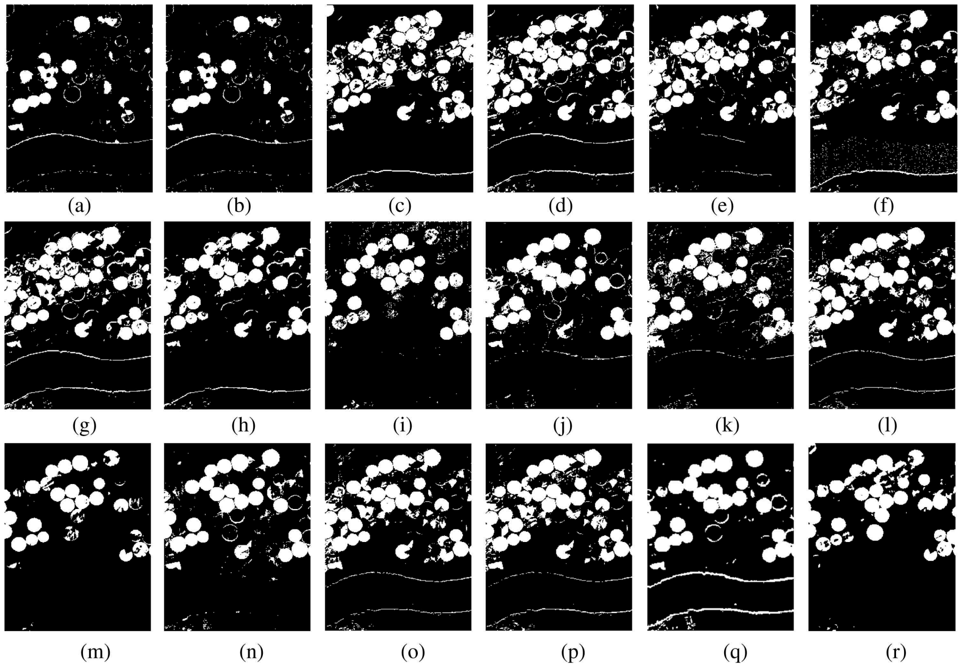


Figure 9. Result of the best performance in G2 for all scenarios in the USA data set using supervised threshold selection: (a) CEN; (b) CEC; (c) ICA; (d) CCA-AD; (e) CE-AD; (f) MCCA-AD; (g) PCA -AD; (h) SVD-AD; (i) SISAL; (j) *N*-finder; (k) LSQ; (l) PCA; (m) MAD; (n) IR-MAD; (o) CE; (p) CRC; (q) SSB; and (r) MLE.

change is necessary. The results presented in Figures 12 and 13 are based on estimating the optimum threshold for each target. To produce good results in change detection, the training data play a key role and should cover all change classes; however, in this study, *no-change* classes are considered as only one class. The best results of the HSCD methods based on visual and numerical analyses among all of the scenarios for the China data set are SVM (S1), MiD (S1), MLC (S1), TCD (S2), SAM (S3), FCM (S1), ISO-Data (S1), and KM (S3). Moreover, for the USA data set they are SVM (S2), MiD (S2), MLC (S1), TCD (S2), SAM (S2), FCM (S1), ISO-Data (S3), and KM (S1). The issue shows that the MiD and MLC methods are robust in different data sets, whereas other techniques have different responses regarding the scenarios and input data sets.

The FCM in G3 has a high variation and therefore the related performance is between 68% and 80%. This issue is due to complexity, atmospheric condition, and noise level in the data set. Among all supervised methods in G3, the SVM algorithm has the best output results and is the most robust one because of achieving a higher accuracy (more than 90%) in all data sets and scenarios.

For the unsupervised methods, it is necessary to assign a primary value to some parameters (e.g. number of clusters) for extracting changes. The number of clusters is a crucial factor in unsupervised change detection. The number of clusters is set to 5 and 8, respectively, for the China and USA data sets. In ISODATA, KM, and FCM methods, the number of iteration is set to 100 for all data sets. In addition, the FCM method used Euclidian metric. The change threshold parameter is set to 35% for all data sets. The FCM

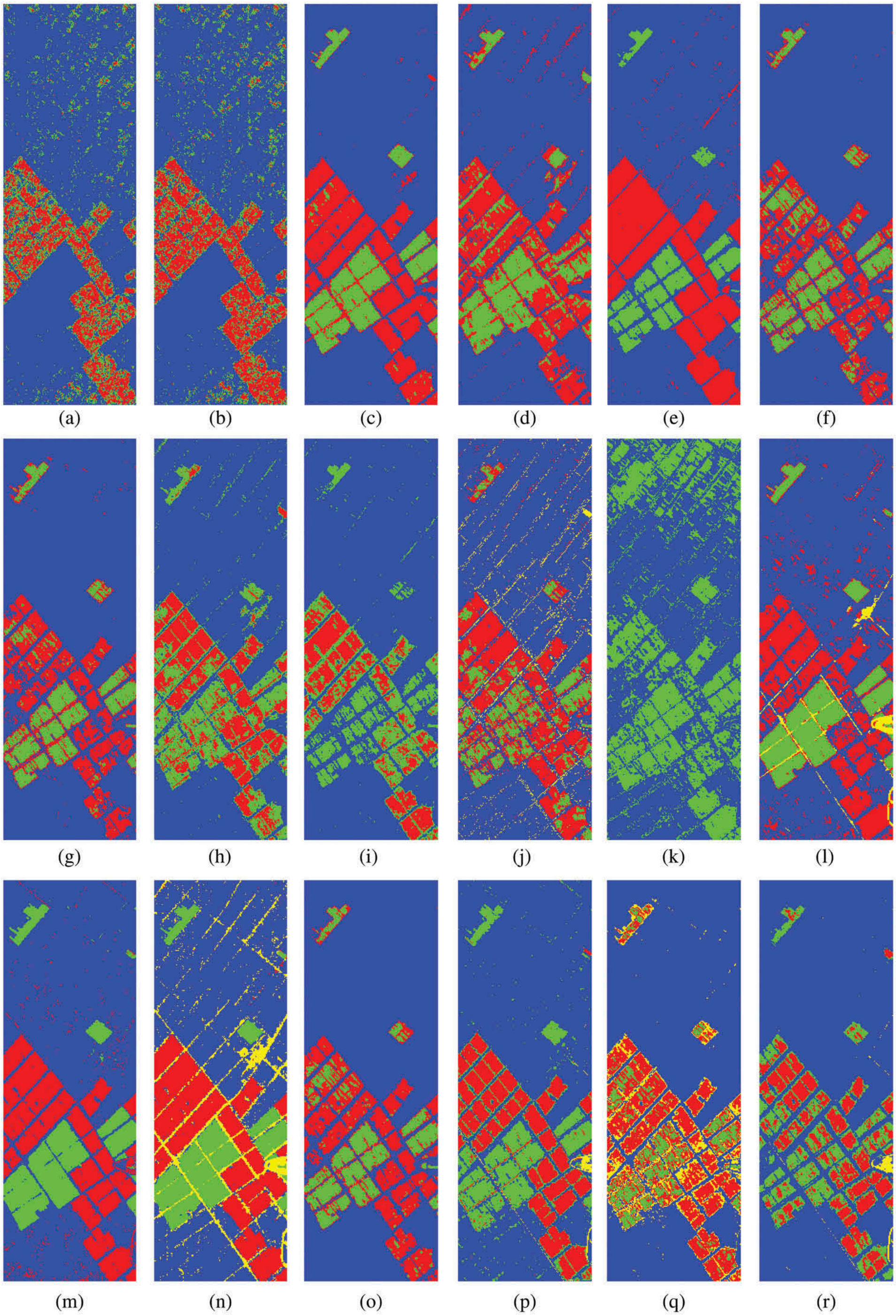


Figure 10. Result of the best performance in G2 for all scenarios in the China data set using unsupervised threshold selection: (a) CEN; (b) CEC; (c) ICA; (d) CCA-AD; (e) CE-AD; (f) MCCA-AD; (g) PCA -AD; (h) SVD-AD; (i) SISAL; (j) *N*-finder; (k) LSQ; (l) PCA; (m) MAD; (n) IR-MAD; (o) CE; (p) CRC; (q) SSB; and (r) MLE.

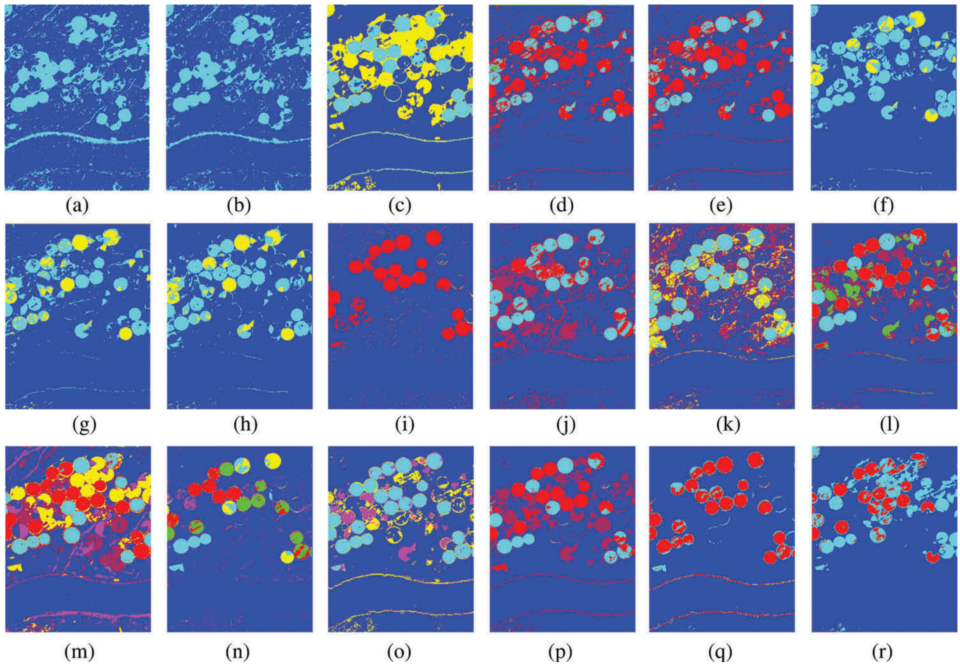


Figure 11. Result of the best performance in G2 for all scenarios in the USA data set using unsupervised threshold selection: (a) CEN; (b) CEC; (c) ICA; (d) CCA-AD; (e) CE-AD; (f) MCCA-AD; (g) PCA -AD; (h) SVD-AD; (i) SISAL; (j) *N*-finder; (k) LSQ; (l) PCA; (m) MAD; (n) IR-MAD; (o) CE; (p) CRC; (q) SSB; and (r) MLE.

method has robust output results in all scenarios and in the two employed data sets. An important issue in unsupervised methods is labelling of the clusters. Each unsupervised method assigns a randomized label for classes. Therefore, it is difficult to find the *change* from *no-change* classes due to different labelling in two data sets. The next issue for methods in G3 is that some main change areas cannot be detected, as it is evident in the China data set.

4.2.4. *Results of the post-classification-based group (G4)*

The experimental results of the post-classification-based methods are shown in Table 13 and Figures 14 and 15 for both USA and China data sets for all of the scenarios. The post-classification-based methods have the same setting parameters compared to the direct classification-based methods, but with different values. For the supervised methods, it is necessary to use training data for each data set. The number of classes in the China data set is constant for multi-date and set to five classes. However, in the USA data set, for t_1 data, the number of classes is set to seven, and for t_2 data, it is set to eight classes.

The SVM algorithm has two main parameters, gamma as kernel parameter set to $\gamma = 0.01$, 0.18 and the penalty coefficient is set to $C = 105$ and 185 for the China and USA data sets, respectively. The 'set max angle' is a parameter (threshold) of the SAM algorithm. If the calculated SAM value for each pixel is greater than the threshold, then the pixel values cannot be classified. Thus, the best threshold for all classes, data sets, and scenarios is set to 0.4. The MLC classifier has two setting parameters; the probability threshold is set to 0.05 for

Table 12. Result performance of G3 for both data sets in all scenarios.

Method	Scenario	Data set	Multiple Change	
			Overall accuracy (%)	κ
FCM	S1	USA	78.00	0.436
		China	79.45	0.654
	S2	USA	72.54	0.333
		China	76.38	0.609
	S3	USA	72.54	0.333
		China	77.86	0.628
	S4	USA	72.54	0.333
		China	67.65	0.506
ISODATA	S1	USA	75.40	0.442
		China	82.50	0.694
	S2	USA	67.44	0.331
		China	82.02	0.681
	S3	USA	71.18	0.369
		China	82.03	0.681
	S4	USA	67.51	0.337
		China	72.72	0.540
KM	S1	USA	72.56	0.397
		China	81.93	0.685
	S2	USA	67.44	0.331
		China	80.05	0.636
	S3	USA	71.18	0.369
		China	82.03	0.681
	S4	USA	67.51	0.337
		China	72.72	0.540
SAM	S1	USA	76.02	0.454
		China	84.31	0.719
	S2	USA	75.08	0.461
		China	82.35	0.687
	S3	USA	73.34	0.444
		China	84.17	0.714
	S4	USA	74.54	0.447
		China	79.55	0.636
SVM	S1	USA	92.91	0.819
		China	94.35	0.883
	S2	USA	92.93	0.820
		China	91.91	0.831
	S3	USA	91.99	0.795
		China	91.93	0.803
	S4	USA	91.89	0.791
		China	89.60	0.779
MiD	S1	USA	71.05	0.405
		China	86.45	0.751
	S2	USA	73.29	0.433
		China	84.06	0.710
	S3	USA	70.78	0.401
		China	85.95	0.740
	S4	USA	72.88	0.424
		China	81.32	0.660
MLC	S1	USA	90.41	0.713
		China	93.18	0.850
	S2	USA	90.35	0.711
		China	89.42	0.762
	S3	USA	88.34	0.648
		China	90.48	0.785
	S4	USA	90.09	0.704
		China	93.18	0.850

(*Continued*)

Table 12. (Continued).

Method	Scenario	Data set	Multiple Change	
			Overall accuracy (%)	κ
TCD	S1	USA	75.37	0.473
		China	81.61	0.667
	S2	USA	78.93	0.513
		China	83.16	0.651
	S3	USA	75.37	0.473
		China	82.65	0.658
	S4	USA	77.73	0.500
		China	76.58	0.539

all classes and the data scale factor is set to the maximum value in each data set. The data set needs to normalize in the range of (0,1) before running the MLC classifier. The MiD algorithm has the maximum distance error that is set to 10^6 and the maximum standard deviation from mean that is set to 10^3. A number of classes (clusters) are available and must be set for ISODATA, FCM, and KM methods. The number of iterations in ISODATA, FCM, and KM is set to 100 for all data sets. In addition, other remaining parameters are set to default values. The change threshold for the KM method is set to 35% for all data sets. One challenge in the unsupervised method is assigning different labels for a scene between two data sets, which must be labelled in a homogenized manner. Moreover, some of the classes are ignored by algorithms and, as a consequence, another class is divided into two clusters. This issue can decrease the performance of HSCD methods and increase the false alarm rates. Based on the results from Table 13 and Figures 14 and 15, it is clear that the best efficiencies based on visual and numerical analyses on multiple changes among the two data sets and in all scenarios are FCM (S3), ISO-Data (S1), KM (S1), SVM (S3), MiD (S1), MLE (S4), and SAM (S1) for the China data set. In addition, the best efficiencies are FCM (S2), ISO-Data (S3), KM (S1), SVM (S2), MiD (S4), MLE (S2), and SAM(S1) for the USA data set. The MLC and SVM algorithms, as compared to the other methods, have a low false alarm rate, whereas the other methods have a high false alarm rate.

The FCM and MiD methods have robust results on both data sets. The unsupervised methods have more variation in different scenarios. Among the unsupervised methods, the best performance in the China data set belongs to FCM with 90.95%, whereas in the USA data set it belongs to KM with 85.28%. Based on the visual analysis of Figure 14, it is clear that all supervised methods have good efficiency for detecting changes and most methods can detect the main changes. The difference in outputs is related to subtle changes in these methods. In the China data set and in the SVM method, the performance is not good. The decrease in the overall accuracy may be related to the false detection originating from mis-registration, presumably.

4.2.5. *Results of the hybrid-based group (G5)*

The experimental results of the hybrid-based methods are presented in Table 14 and Figures 16 and 17 for both USA and China data sets among all scenarios with the supervised threshold selection. The unsupervised framework is dominated by hybrid methods. These methods have a variety of setting parameters compared to the other groups. The C^2VA method has three parameters: alpha is between 0.51 and 0.69, the number of iterations is 15, and the difference to reach to the exit EM loop is set to 10^{-6}. In

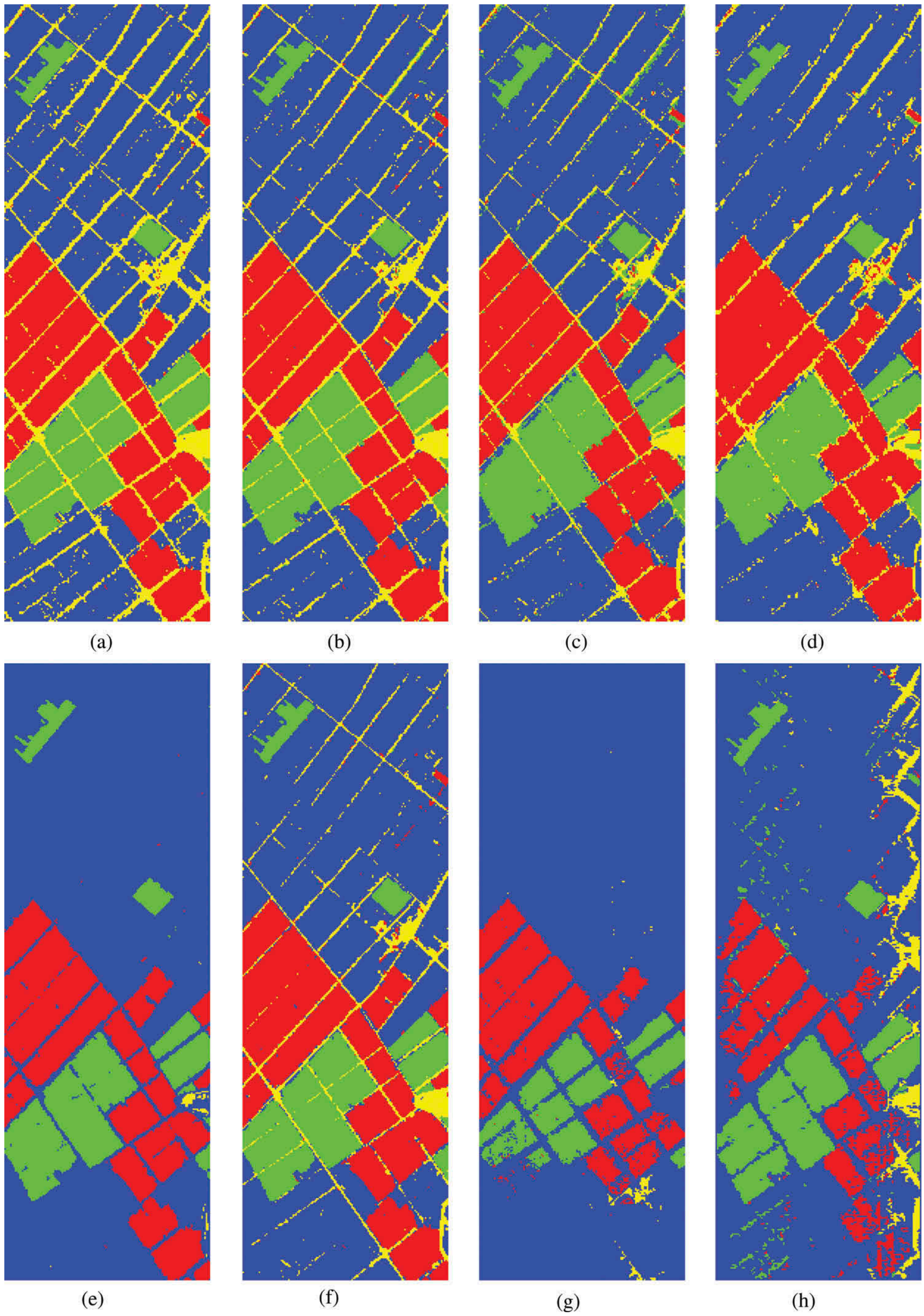


Figure 12. Result of the best performance in G3 for all scenarios in the China data set: (a) FCM; (b) ISODATA; (c) KM; (d) SAM; (e) SVM; (f) MiD; (g) MLE; and (h) TCD.

SCRC, the number of segments is set to 5 and 7 for the China and USA data sets, respectively. In addition, segmentation is applied to the first time data set. In SPIM, the number of clusters is 3, the number of iterations is 100, initial weights is 1, epsilon is 10^{-2},

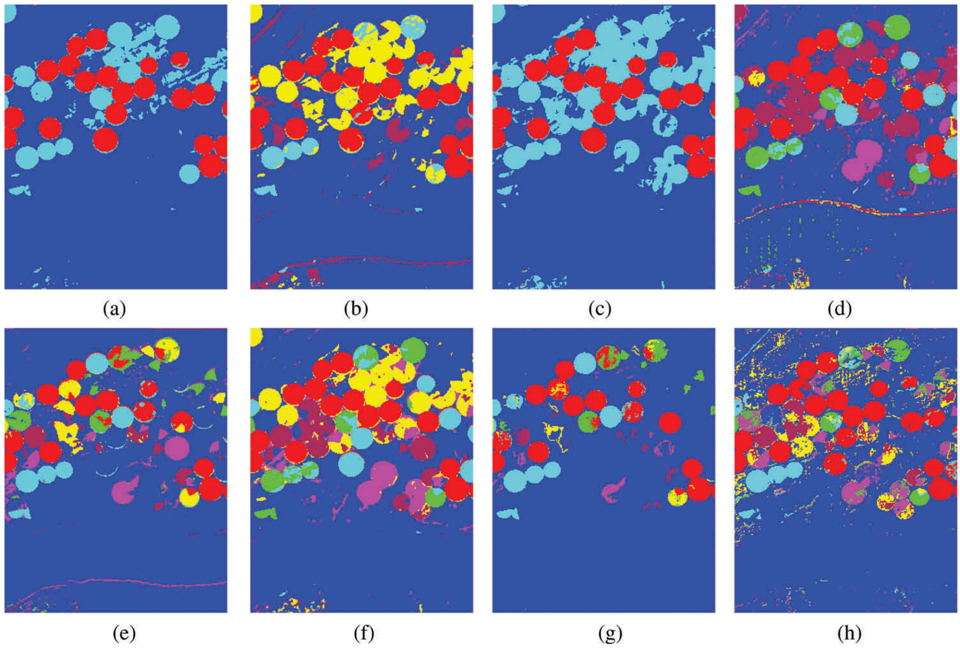


Figure 13. Result of the best performance in G3 for all scenarios in the USA data set: (a) FCM; (b) ISODATA; (c) KM; (d) SAM; (e) SVM; (f) MiD; (g) MLE; and (h) TCD.

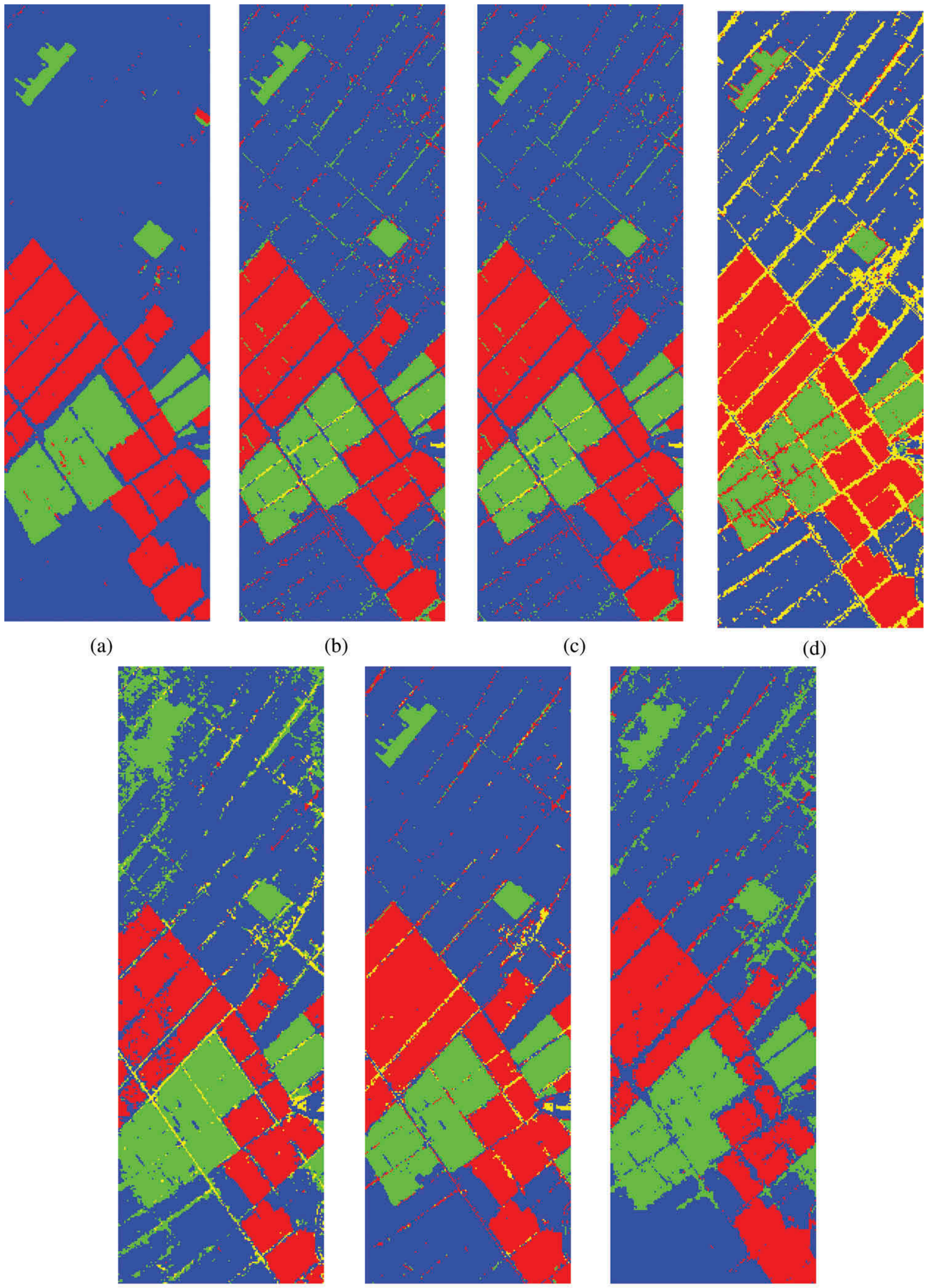
and the used number of incorporated PCs in PCA is 3. Moreover, in the MSU method, alpha is 0.4, the number of iterations is 15, and the difference to reach the exit EM loop is 10^{-6}. Similarly, the threshold in the China and USA data sets is 0.04 and 0.1, respectively, the false rate is 10^{-4}, and the number of the patch is 4. The SCM-CD methods were applied in two scenarios – SCM-CD-SVM and SCM-CD-Threshold. The SCM-CD method has two parameters regularization $\alpha_v = 0.8$, $\alpha_\zeta = 0.55$ By combining this method with SVM, kernel parameter (γ=10^{-3}, 10^{-5}) and C (10^{3}, 10^{5}) also added to previous parameters. In addition, the SCM-CD-threshold method applied with the *k*-nearest neighbour (*k*-NN) algorithm which *k* set to 4 and distance metric measure is ED..

Owing to utilizing segmentation for change detection, the SCRC method is expected to have a good performance compared to the CRC method. Unexpectedly, on change detection, the SCRC method could not outperform the CRC method. The reason for this is that SCRC overestimates the value of some *no-change* areas in the foreground. Therefore, local selection of a suitable threshold is necessary for the SCRC method. The SPIM method is more complicated because the PCA method predicted the *change* areas with positive and negative values (high intensity) and the *no-change* area has values near zero (low intensity). Since the number of clusters is set to 3 for extracting main changes, some main changes could not be extracted by the KM method and those mixed with the *no-change* class. This issue is due to the procedure of the KM algorithm that considered the first cluster as *change* although those are *no-changes*. Therefore, this issue can decrease the performance of change detection in this method. However, the MSU method is based on unmixing and can improve the performance of endmember extraction on the patch-based procedure. On the other hand, there are some issues for merging and

Table 13. Result performance of G4 for both data sets in all scenarios.

Method	Scenario	Data set	Multiple change	
			Overall accuracy (%)	κ
FCM	S1	USA	78.58	0.496
		China	79.19	0.649
	S2	USA	78.74	0.383
		China	72.49	0.535
	S3	USA	78.15	0.479
		China	90.95	0.816
	S4	USA	66.09	0.307
		China	90.72	0.816
ISODATA	S1	USA	69.82	0.321
		China	89.19	0.789
	S2	USA	81.10	0.496
		China	78.03	0.610
	S3	USA	83.33	0.592
		China	78.80	0.624
	S4	USA	77.21	0.460
		China	67.88	0.469
KM	S1	USA	85.28	0.545
		China	89.19	0.789
	S2	USA	85.15	0.660
		China	77.56	0.603
	S3	USA	77.14	0.523
		China	78.80	0.624
	S4	USA	77.26	0.460
		China	67.88	0.468
SAM	S1	USA	84.74	0.542
		China	80.58	0.665
	S2	USA	81.14	0.531
		China	76.88	0.603
	S3	USA	81.54	0.536
		China	79.16	0.628
	S4	USA	79.01	0.478
		China	71.03	0.529
SVM	S1	USA	85.5	0.655
		China	81.64	0.671
	S2	USA	85.96	0.666
		China	79.32	0.624
	S3	USA	82.94	0.612
		China	82.58	0.681
	S4	USA	84.15	0.618
		China	80.00	0.645
MiD	S1	USA	83.22	0.594
		China	88.52	0.779
	S2	USA	84.26	0.620
		China	85.37	0.724
	S3	USA	81.99	0.594
		China	85.71	0.731
	S4	USA	58.04	0.607
		China	85.94	0.732
MLC	S1	USA	84.70	0.650
		China	76.24	0.581
	S2	USA	85.15	0.660
		China	80.85	0.648
	S3	USA	77.14	0.523
		China	83.14	0.688
	S4	USA	78.92	0.532
		China	83.41	0.690

determining endmembers in the *change* and *no-change* areas, which can affect HSCD. The SCM-CD method has a good performance, which is the best performance related to SCM-CD-SVM, but it needs more time for estimating and tuning the setting parameters. Three



(a) (b) (c) (d) (e) (f) (g)

Figure 14. Result of the best performance in G4 for all scenarios in the China data set: (a) FCM; (b) ISODATA; (c) KM; (d) SAM; (e) SVM; (f) MiD; and (g) MLE.

methods of SCRC, SPIM, and ICA-ID need to select a threshold and the threshold selection should be applied similarly to match-based and transformation-based groups. In addition, threshold selection is a built-in section of the SSDM-CD method.

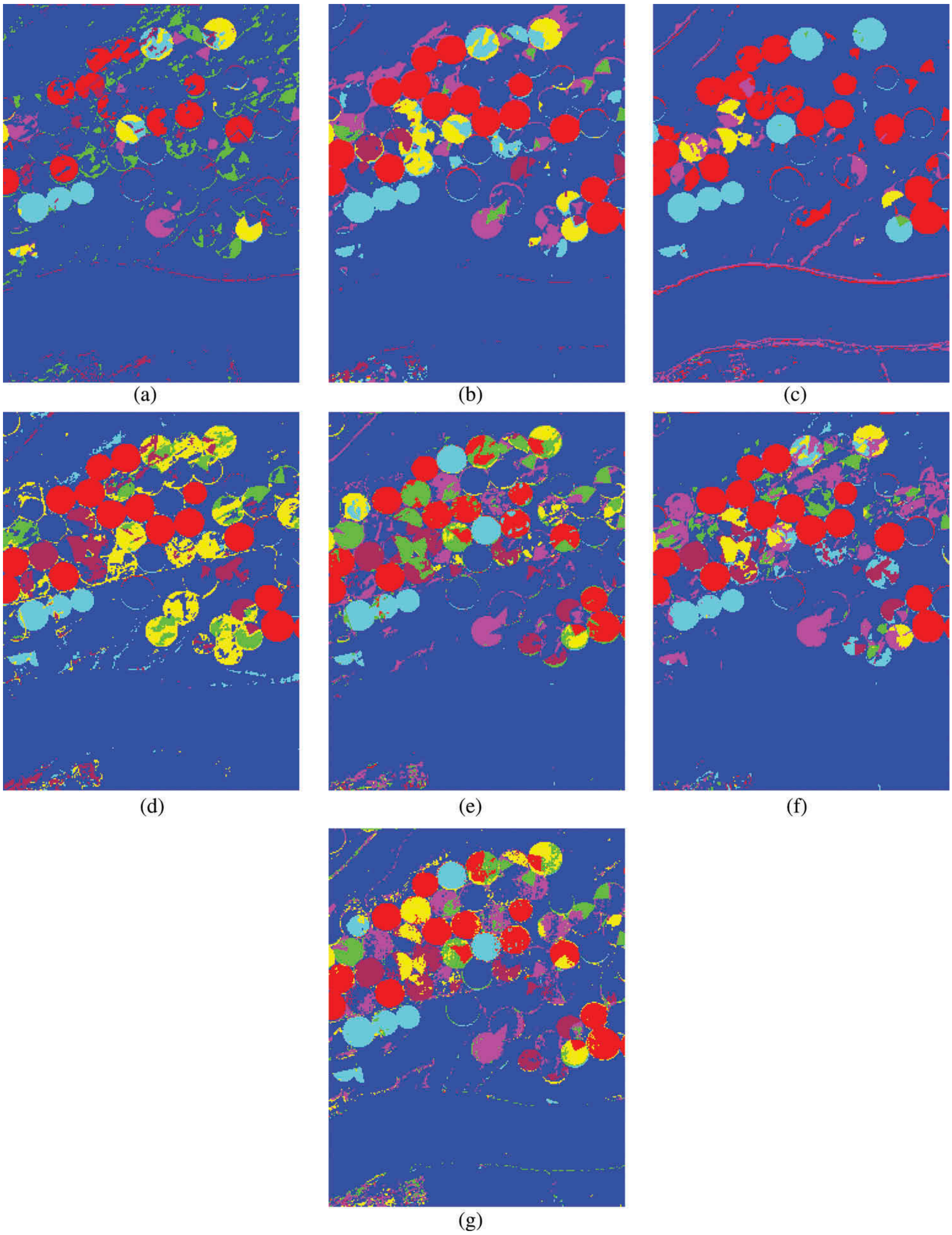


Figure 15. Result of the best performance in G4 for all scenarios in the USA data set: (a) FCM; (b) ISODATA; (c) KM; (d) SAM; (e) SVM; (f) MiD; and (g) MLE.

The results of C^2VA and MSU are multiple change maps in an unsupervised framework. However, the results were converted into binary change maps to be comparable to other methods of this group. The best efficiencies in G5 based on visual and numerical analyses for all scenarios and for the China data set in the state of supervised threshold selection are SCRC (S1), SPIM (S1), ICA-ID (S1), SSDM-CD-SVM(S1), and SSDM-CD-Threshold (S4). In addition, for the USA data set, the best performances are SCRC

Table 14. Result performance of G5 for both data sets in all scenarios.

			Overall accuracy (%)		κ		
Method	Scenario	Data set	Supervised	Unsupervised	Supervised	Unsupervised	AUC
SCRC	S1	USA	94.03	90.69	0.825	0.744	0.964
		China	89.75	88.72	0.770	0.737	0.925
	S2	USA	93.18	88.98	0.791	0.674	0.965
		China	89.62	87.44	0.765	0.711	0.923
	S3	USA	90.68	85.79	0.700	0.586	0.911
		China	89.56	88.17	0.766	0.712	0.927
	S4	USA	91.96	87.12	0.754	0.627	0.938
		China	89.46	87.06	0.764	0.739	0.925
SPIM	S1	USA	89.88	87.37	0.698	0.659	0.964
		China	94.07	83.11	0.862	0.648	0.979
	S2	USA	92.83	90.59	0.775	0.722	0.940
		China	92.20	82.69	0.817	0.626	0.967
	S3	USA	86.96	82.79	0.571	0.489	0.965
		China	89.5	91.6	0.754	0.811	0.945
	S4	USA	92.07	91.19	0.748	0.736	0.938
		China	89.71	83.71	0.758	0.645	0.945
ICA-ID	S1	USA	90.61	87.69	0.7115	0.603	0.931
		China	97.44	91.524	0.940	0.796	0.985
	S2	USA	88.56	88.65	0.641	0.628	0.920
		China	90.13	82.22	0.766	0.599	0.926
	S3	USA	90.68	84.10	0.700	0.588	0.888
		China	91.31	92.12	0.796	0.820	0.936
	S4	USA	91.17	93.29	0.739	0.805	0.926
		China	93.49	91.13	0.848	0.793	0.974
C^{2}VA	S1	USA	87.46	– – – –	– – – –	0.602	– – – –
		China	96.57	– – – –	– – – –	0.921	– – – –
	S2	USA	97.42	– – – –	– – – –	0.928	– – – –
		China	86.99	– – – –	– – – –	0.719	– – – –
	S3	USA	92.82	– – – –	– – – –	0.807	– – – –
		China	89.32	– – – –	– – – –	0.766	– – – –
	S4	USA	94.59	– – – –	– – – –	0.838	– – – –
		China	84.38	– – – –	– – – –	0.676	– – – –
MSU	S1	USA	91.80	– – – –	– – – –	0.774	– – – –
		China	93.96	– – – –	– – – –	0.864	– – – –
	S2	USA	86.72	– – – –	– – – –	0.637	– – – –
		China	94.14	– – – –	– – – –	0.884	– – – –
	S3	USA	90.30	– – – –	– – – –	0.759	– – – –
		China	91.71	– – – –	– – – –	0.814	– – – –
	S4	USA	94.65	– – – –	– – – –	0.852	– – – –
		China	95.95	– – – –	– – – –	0.889	– – – –
SSDM-CD-SVM	S1	USA	93.32	– – – –	0.820	– – – –	– – – –
		China	97.54	– – – –	0.943	– – – –	– – – –
	S2	USA	93.30	– – – –	0.819	– – – –	– – – –
		China	94.56	– – – –	0.874	– – – –	– – – –
	S3	USA	92.85	– – – –	0.805	– – – –	– – – –
		China	94.80	– – – –	0.879	– – – –	– – – –
	S4	USA	93.54	– – – –	0.823	– – – –	– – – –
		China	94.75	– – – –	0.880	– – – –	– – – –
SSD-CD-Threshold	S1	USA	91.4	– – – –	0.772	– – – –	– – – –
		China	92.97	– – – –	0.838	– – – –	– – – –
	S2	USA	91.63	– – – –	0.776	– – – –	– – – –
		China	92.48	– – – –	0.828	– – – –	– – – –
	S3	USA	92.12	– – – –	0.788	– – – –	– – – –
		China	92.97	– – – –	0.838	– – – –	– – – –
	S4	USA	91.57	– – – –	0.774	– – – –	– – – –
		China	93.59	– – – –	0.853	– – – –	– – – –

(S1), SPIM (S2), ICA-ID (S4), SSDM-CD-SVM(S4), and SSDM-CD-Threshold (S3). In G5, the most robust methods are MSU, C^{2}VA, and SCM-CD methods.

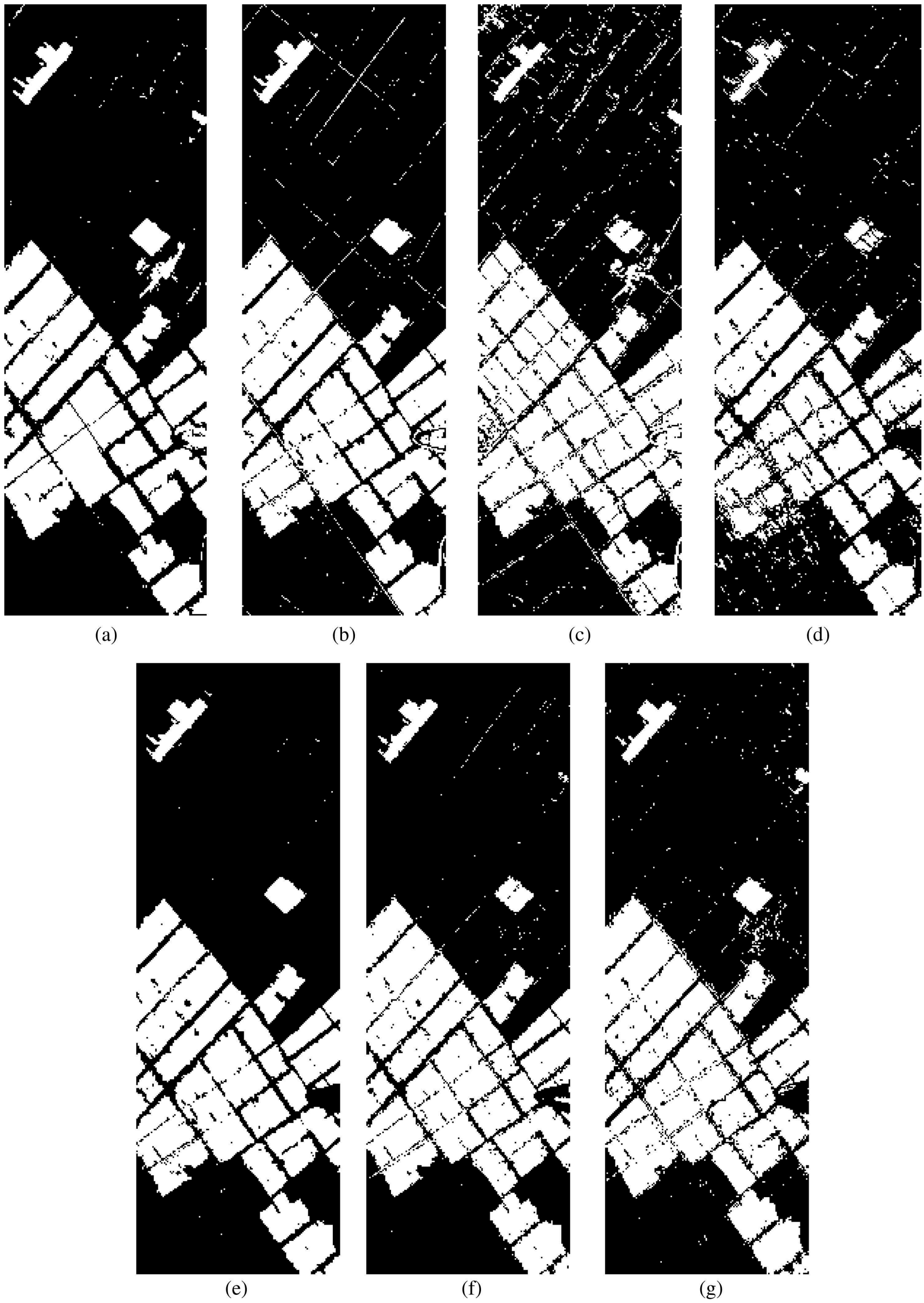


Figure 16. Result of the best performance in G5 for all scenarios in the China data set: (a) MSU; (b) C^2VA; (c) SCRC; (d) SPIM; (e) ICA-ID; (f) SSDM-CD-SVM; and (g) SSDM-CD-Threshold.

Figures 18 and 19 and Table 14 present the results of multiple change classes. The best performances for the China data set among the four scenarios are SCRC (S1), SPIM (S3), ICA-ID (S1), C^2VA (S1), and MSU (S4); also for the USA data set, the results are SCRC

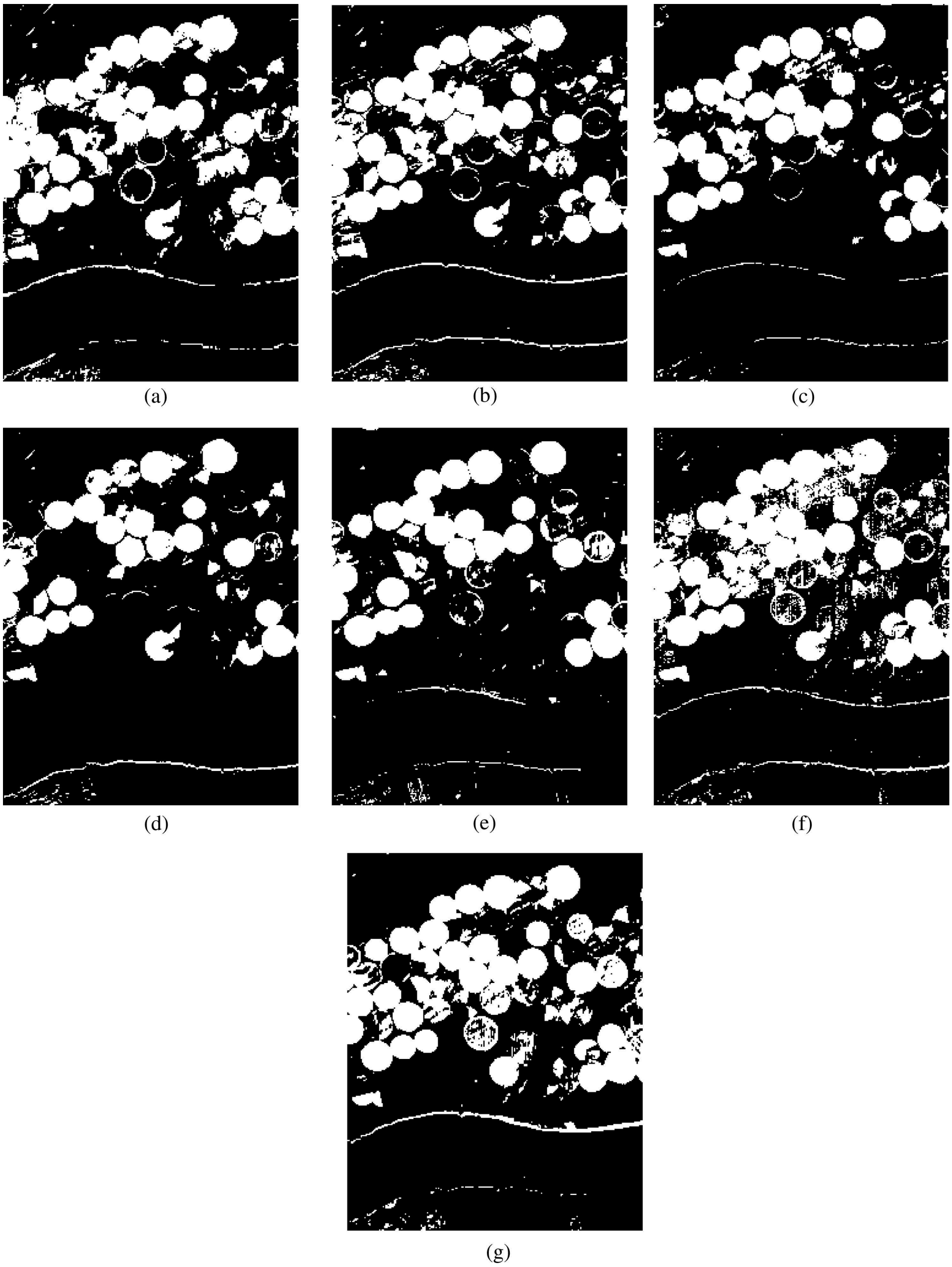


Figure 17. Result of the best performance in G5 for all scenarios in the USA data set: (a) MSU; (b) C^2VA; (c) SCRC; (d) SPIM; (e) ICA-ID; (f) SSDM-CD-SVM; and (g) SSDM-CD-Threshold.

(S1), SPIM (S4), ICA-ID (S4), C^2VA (S2), and MSU (S4). Based on the results illustrated in Table 14, some of the scenarios show that the unsupervised thresholding improves the performance of HSCD. The best thresholding performance among the three methods is that of ICA-ID for both data sets. The numbers of *change* classes detected by

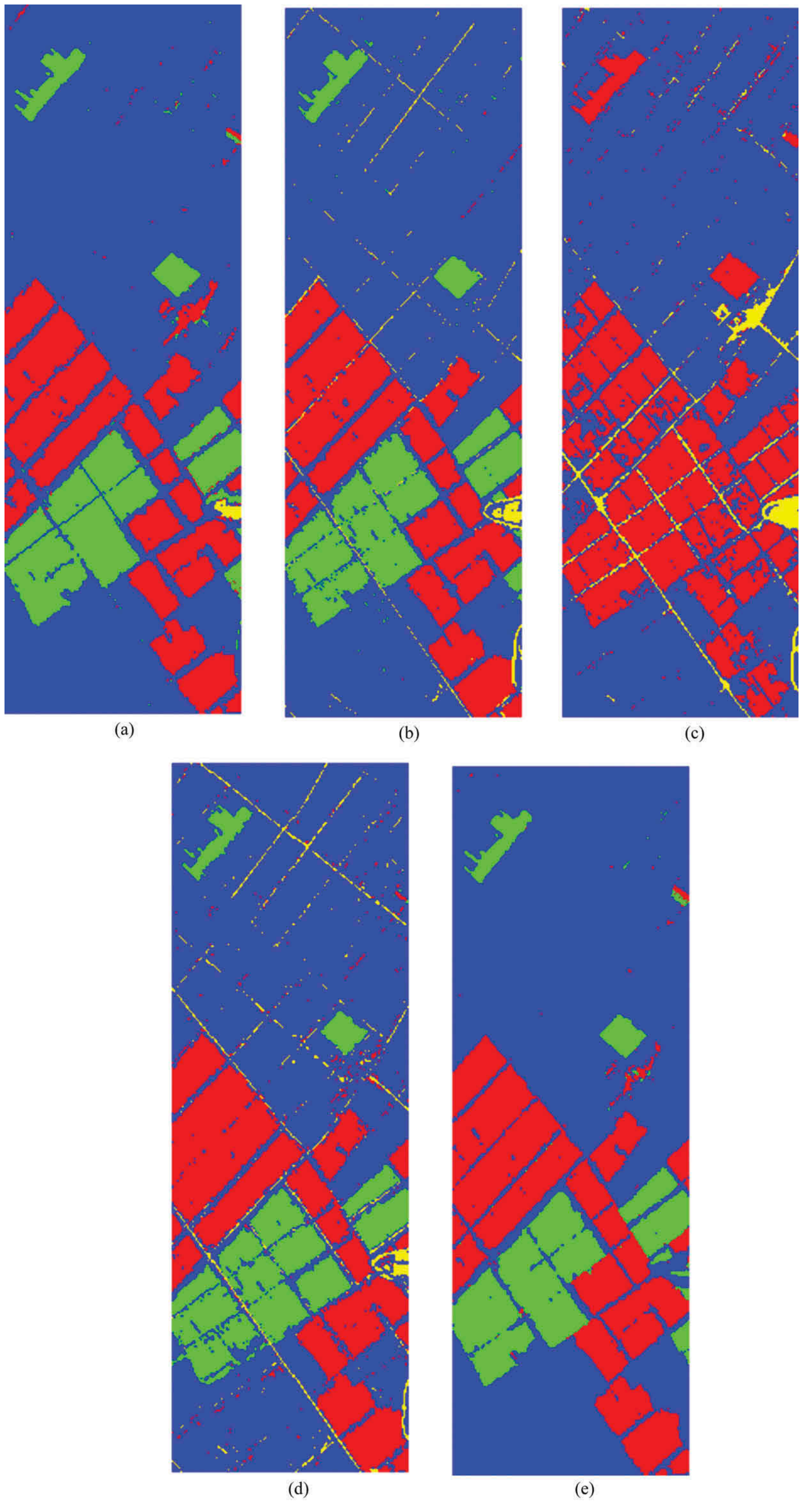


Figure 18. Result of the best performance in G5 for all scenarios in the China data set using unsupervised threshold selection: (a) MSU; (b) C^{2}VA; (c) SCRC; (d) SPIM; and (e) ICA-ID.

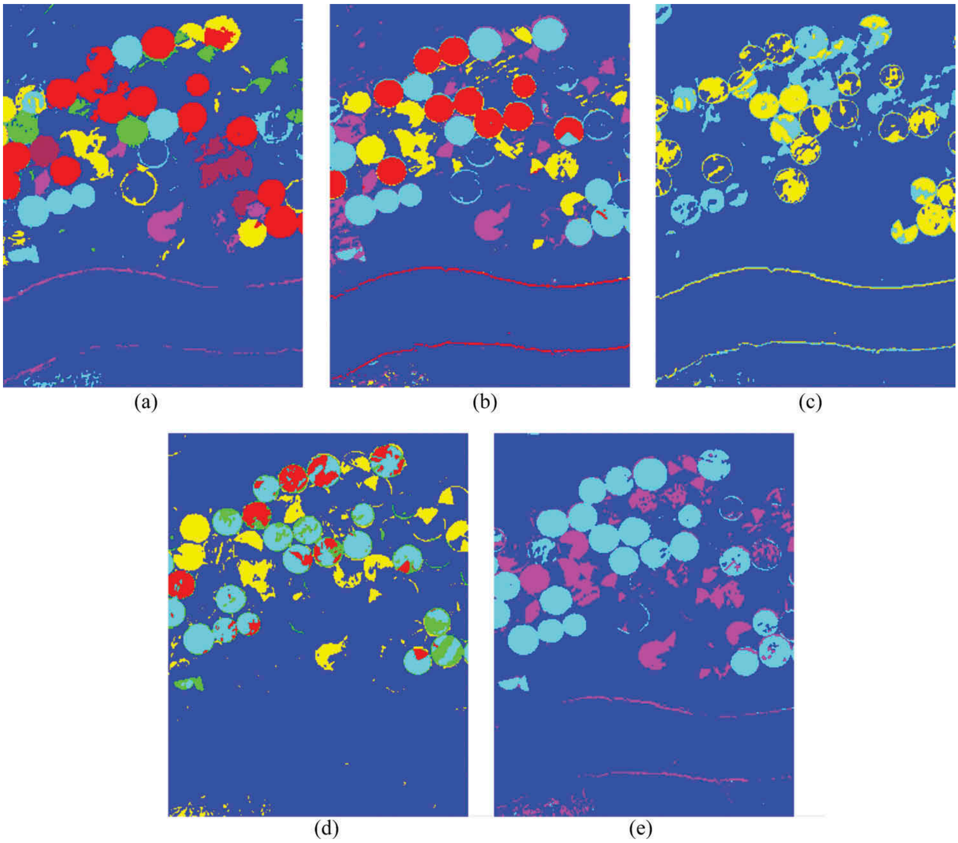


Figure 19. Result of the best performance in G5 for all scenarios in the USA data set using unsupervised threshold selection: (a) MSU; (b) C^2VA; (c) SCRC; (d) SPIM; and (e) ICA-ID.

unsupervised methods for the multiple change map in the China data set are as follows: SCRC (ω_1, ω_3), SPIM (ω_1, ω_2, ω_3), ICA-ID (ω_1, ω_2), C^2VA (ω_1, ω_2, ω_3), and MSU (ω_1, ω_2, ω_3). In addition, the multiple change class results, using multiple thresholding for the USA data set, are as follows: SCRC (ω_3, ω_4), SPIM (ω_1, ω_2, ω_3, ω_4), ICA-ID (ω_4, ω_3), C^2VA (ω_1, ω_3, ω_4, ω_5), and MSU (ω_1, ω_2, ω_3, ω_4, ω_5, ω_6).

5. Discussion

5.1. *The strengths and weaknesses of the match-based groups (G1)*

In Table 15, the noise, atmospheric condition, implication, threshold-based methodology, and other characteristics of the match-based methods are discussed. These results reveal that match-based methods have common properties (Table 15), including the following.

(1) These methods have better performance in comparison with other groups and also the overall accuracy for most of these methods is higher than 90%.

Table 15. Characterization of G#1 methods.

Method	Noise	Atmospheric	Other
SAM	Yes	No	Low accuracy in complex area and water area, the accuracy depends on ground truth, problem is finding optimal threshold
ID	No	No	The accuracy depends on ground truth, problem is finding optimal threshold, need two thresholds for CD
IR	No	No	The accuracy depends on ground truth, problem is finding optimal threshold, need two thresholds
CC	No	Yes	High accuracy in complex area, the accuracy depends on ground truth, problem is finding optimal threshold
MD	Yes	Yes	The accuracy depends on ground truth, problem is finding optimal threshold, need two thresholds, lowest accuracy in SB methods
ED	No	Yes	The accuracy depends on ground truth, problem is finding optimal threshold

(2) They have a simpler implementation compared to other methods because they use simple mathematical operators.
(3) They require lower processing time.
(4) The structure of incorporated data is cube–cube; therefore, all spectral bands are used for change detection, so in these groups finding the optimal bands for change detection is not necessary.
(5) They do not need parameter setting.
(6) These methods do not provide ‘from-to’ information.
(7) The main problem with these methods is finding the optimal threshold.

The similarity-based methods show a better performance compared to the match-based methods, but their sensitivity to preprocessing and accuracy variation is high in the different covered scenarios. In simple areas with the same land-cover type, the SAM methods have high efficiency, whereas in complex areas, CC and ED perform better. In sum, the distance-based methods show a lower sensitivity to preprocessing and a higher average performance accuracy.

Among all of the method groups (G1), the ID and IR methods focused on multiple changes more than the other methods. Therefore, for multiple change detection, both of them are suitable. Some methods, such as CCO, ED, and SAM, are suitable for binary change map, meaning that the output of these methods is a single band. Therefore, due to the lack of a spectral component, finding a multiple change map is difficult and is thus inefficient in multiple thresholding methods. These methods are based on the cube–cube structure and the classifier algorithm could not extract more information about multiple changes. In sum, algorithms in this group focused on the measurement changes based on the spectral profile. Therefore, due to incorporating different time period data sets, the high systematic errors (noise and striping) of the sensor, and environment conditions, it is better not to use the algorithms of this group for reducing high false alarms.

5.2. *The strengths and weaknesses of the transformation-based groups (G2)*

The transformation-based methods were investigated similar to match-based methods by evaluating the sensitivity to noise, atmospheric condition, time of processing,

Table 16. Characterization of G#2 methods.

Method	Time	Implication	Structure	Threshold selection	Having a criterion for optimal bands	Sensitive to predicted type of changes
CEN	low	easy	cube–cube	yes	– – – – –	very low
CEC	low	easy	cube–cube	yes	– – – – –	very low
ICA	Medium	medium	band–band	yes	No	medium
CRC-SVD-AD	low	medium	band–cube	yes	– – – – –	low
CE-AD	low	medium	band–cube	yes	– – – – –	low
MCC-AD	low	medium	band–cube	yes	– – – – –	very low
PCA-AD	low	medium	band–cube	yes	– – – – –	low
SVD-AD	medium	medium	band–cube	yes	– – – – –	low
SISAL	medium	medium	cube–cube	yes	– – – – –	medium
N-finder	medium	medium	cube–cube	yes	– – – – –	medium
LSQ	medium	medium	cube–cube	yes	– – – – –	medium
PCA	low	easy	band–band	yes	yes	high
MAD	low	easy	band–band	yes	no	medium
IR-MAD	medium	medium	band–band	yes	no	high
CC	low	easy	band–band	yes	no	high
CE	low	easy	band–band	yes	no	high
SSB	low	easy	cube–cube	yes	– – – – –	medium
MLE	low	easy	cube–cube	yes	– – – – –	medium

implication, 'from-to' information, data usage structure, threshold-based methods, and other characteristics. The output of the performance of the transformation-based methods, as well as an abstract of this group with a specific characteristic, is illustrated in Table 16. These methods are divided into three categories: the general methods, the unmixing methods, and the combined transformation-based and AD method.

The experimental result of general transformation-based methods is summarized as follows. (1) Most of the methods in this group enable noise control in their procedures. (2) Some methods have criteria for evaluating each band (such as PCA that benefits from eigenvalues for sorting bands), which is helpful for selecting the optimum bands faster. Moreover, dimensionality reduction is necessary for most of the applications in the combined methods. (3) Finally, some methods have low sensitivity to atmospheric condition. The CE, CRC, MAD, IR-MAD, and PCA methods have better performance in this category. These methods do not require parameter setting except the IR-MAD. Most of the methods in this group (general TB methods) have good performance on multiple change detection due to producing negative and positive change values.

The result of the unmixing transformation-based methods shows good efficiency and can provide more details about the changes. Moreover, except for the initial parameters, these methods do not require any parameter settings. Nevertheless, in these methods, the robust preprocessing for extracting the changes is inevitable. This is due to noise and its subsequent effect on the unmixing analysis, as well as the fact that computing an abundance map for the complicated land-cover area is difficult and time-consuming. The SISAL and *N*-finder methods do not have outputs in S3 because the HySime algorithm does not provide any estimation.

We provide a summary of our experimental results of the combined transformation-based and AD algorithms as follows.

(1) In this approach, it is not necessary to find the optimum bands.
(2) In these methods, it is necessary to select informative bands due to incorporation of AD methods that are affected by noisy bands, and detect these noises as changes.
(3) The systematic errors (noise and de-striping) among the components' predictor algorithms are extracted as changes.
(4) Due to using the AD algorithm, as a consequence all components covert to a band and as a result the finding of the multiple changes is difficult.
(5) These methods do not require setting parameters. Furthermore, among all three categories of the transformation-based methods, this category has the lowest accuracy on change detection.

5.3. *The strengths and weaknesses of the direct classification-based groups (G3)*

For more details on the results of direct classification-based methods, Table 17 is composed. We can summarize our experimental results on the direct classification-based methods as follows: (1) they show little sensitivity to noise and atmospheric conditions, (2) their misclassification error is low, (3) they need high-performance computer systems for processing due to the stacking approach, (4) they have more accuracy compared to post-classification-based methods, and (5) they require various parameter settings. Owing to high dimensional stacked data and time-consuming processing, performing the dimensionality reduction techniques in direct classification-based methods is a big challenge. The unsupervised direct classification-based methods can be summarized as follows: (1) they need more time for classification, (2) estimating the number of classes is a challenge; and (3) estimating and tuning of parameters for the classifiers are necessary and the unsupervised method requires more parameters than the supervised direct classification-based methods. Similar to the supervised post-classification-based methods, the unsupervised direct classification-based methods need dimensionality reduction and training data, but the misclassification error is lower. One of the most important characteristics of the direct classification-based methods compared to the post-classification-based methods is considering all *no-change* classes as one class.

5.4. *Strengths and weaknesses of the post-classification-based groups (G4)*

A discussion on the properties and characteristics of post-classification-based methods is presented in Table 18. The visual analysis and numerical results show that the post-classification-based methods have common characteristics.

(1) The post-classification-based methods have less sensitivity to noise and atmospheric conditions.
(2) They show a high overall accuracy in HSCD methods on both data sets.
(3) They can provide full 'from-to' information compared to other groups.
(4) Their misclassification error is low.
(5) They require different parameter settings.

The unsupervised post-classification-based methods provide a high accuracy and do not need training data, but they suffer from a number of issues, such as

Table 17. Characterization of G#3 methods.

Method	Noise	Atmospheric	Time	Implication	'from-to'	Multiple Change	Mis-classification	Label problem	Other
KM	no	no	high	medium	no	yes	medium	yes	Need estimation of initial of setting parameters and clusters.
FCM	very low	very low	high	medium	no	yes	medium	yes	Misclassification, the problem is a lack of the same label for classes, estimation of initial of setting parameters and clusters.
ISODATA	very low	very low	high	medium	no	yes	medium	yes	Need estimation of initial of setting parameters and clusters.
TCD	low	low	medium	medium	low	no	low	– – – –	Need dimensional reduction, accuracy depends on the quality of training data, difficult to provide training data, need threshold selection for each class.
SAM	low	low	low	easy	low	no	low	– – – –	Need dimensional reduction, train-set output depends on quality of training data, sensitive to complex data.
SVM	very low	no	medium	medium	low	no	very low	– – – –	Need dimensional reduction, result depends on the quality of training data, optimizing kernel parameters and SVM parameters.
MLC	low	no	low	easy	low	no	very low	– – – –	Need dimensional reduction and train-set output depends on quality of training data.
MiD	very low	no	low	easy	low	no	low	– – – – -	Have false alarm rate, need dimensional reduction, train-set output depends on quality of training data.

Table 18. Characterization of G#4 methods.

Method	Noise	Atmospheric	Time	Implication	'from-to'	Multiple change	Mis-classification	Label problem	Other
KM	medium	very low	high	medium	no	yes	medium	yes	Need estimation of initial of setting parameters and clusters.
FCM	very low	low	high	medium	no	yes	medium	yes	Need estimation of initial of setting parameters and clusters
ISODATA	low	low	high	medium	no	yes	medium	yes	Need estimation of initial of setting parameters and clusters
SAM	low	no	medium	easy	yes	no	low	– – – –	Need training data and dimensional reduction, output depends on the quality of training data
SVM	very low	very low	low	medium	yes	no	very low	– – – –	Need dimensional reduction, output depends on the quality of training data, difficult optimizing kernel parameters and SVM parameters
MLC	low	low	low	easy	yes	no	low	– – – –	Need training data and dimensional reduction, output depends on the quality of training data
MiD	medium	low	low	easy	yes	no	low	– – – –	Need training data and dimensional reduction, output depends on the quality of training data

(1) time-consuming processing;
(2) because of a mismatch between the entity of each class and its label (which is the result of random labelling), the supervised labelling for change detection is necessary; and
(3) estimating the number of clusters and other parameters is necessary.

On the one hand, for the supervised methods, processing does not take much time, implicating that they are not very complex, and therefore they can improve the accuracy by performing the transformation from the image space to the feature space. However, these methods suffer from a series of drawbacks, including (1) sensitivity to the number and quality of training data; (2) dimensionality reduction is inevitable due to the Hughes phenomenon; and (3) in the SVM classifier, it is necessary to select the kernel and other parameters. The collection of training data is also a big challenge when dealing with the multi-temporal data sets. It is noteworthy that the FCM method displays a better accuracy compared to the other methods in this group.

5.5. *Strengths and weaknesses of the hybrid-based groups (G5)*

Table 19 provides the details of the properties and characteristics of hybrid-based methods in more detail. It is clear from this table that the hybrid-based methods show better accuracy compared to the previous groups and the ability to control the effects of noise and atmospheric conditions. For this group (G5), we can mention that most of the methods: (1) can provide multiple change information; (2) can provide a better overall performance compared to the other groups; and (3) require a parameter setting. It is worth pointing out that the MSU method provides the best accuracy in this group among the unsupervised algorithms, but estimating the parameters demands prior knowledge. Additionally, the SSDM-CD as the semi-supervised method has a good efficiency for detecting changes, but training data and tuning parameters are an integral part of this method. The MSU, C^2VA, and SPIM methods focus on the multiple change map and have a high potential for detecting multiple changes. The MSU method has the best performance on the multiple change map and all *change* classes obtained by it are meaningful.

6. Conclusion

In this study, we provided an experimental comparative evaluation of different HSCD methods using two real-world publicly available hyperspectral images. This evaluation covered important parameters such as sensitivity to noise, atmospheric condition, time of processing, the structure of the algorithm, 'from-to' and 'multiple-change' information, and threshold selection. We discussed the advantages and disadvantages of each group of algorithms based on experimental results. For this purpose, all HSCD methods were categorized into five groups – match-based, transformation-based, post-classification-based, direct classification-based, and hybrid-based groups. Also as part of this study, four scenarios were designed to monitor the effects of spatial, atmospheric, and radiometric conditions and noise reduction using all HSCD methods in those five groups. Based on the obtained results, the accuracy of many HSCD methods depends on several factors. (1) Preprocessing (quality enhancement): it includes spatial (geometric correction) and spectral (de-noising, de-striping, atmospheric correction, radiometric conditions)

Table 19. Characterization of G#5 methods.

Method	Noise	Atmospheric	Time	Implication	Multiple change	Structure	Threshold	Other
SCRC	very low	very low	medium	medium	no	band–band	yes	Need to estimate the number of classes, the problem is finding optimal band and thresholds.
ICA-ID	low	no	medium	medium	no	band–band	yes	The problem is finding optimal band and thresholds.
C^2VA	low	low	medium	medium	yes	cube–cube	no	The accuracy depends on EM clustering, EM needs initial parameters.
MSU	low	low	medium	medium	yes	cube–cube	no	Using many algorithms, based on a knowledge-based threshold and focus on multiple change map.
SPIM	low	low	medium	easy	no	cube–cube	no	Problem is extracting uncertain changes using clustering.
SSD-CD	low	very low	medium	medium	no	cube–cube	no	Need training data, based on a knowledge-based threshold.

correction. Therefore, the accuracy of the HSCD methods can be improved using the novel spatial and spectral correction techniques. (2) For transformation-based and match-based methods, two factors play a key role in performance and accuracy, including, first, the ability to correctly predict the type of land-cover *change* from *no-change* areas by the change detection method and, second, the threshold selection method. Of course, the precise prediction of *change* and *no-change* areas leads to creating an apparent histogram from *change* and *no-change* areas, i.e. using the thresholding method could give a higher accuracy to change detection. (3) The quality of collecting training data is important for direct-classification, post-classification, and semi-supervised methods. It should cover all of the classes. (4) For some methods, the setting parameters have a significant impact on accuracy. Thus, to achieve a higher accuracy, it is necessary to determine the optimum value. For this purpose, incorporating the optimization algorithm could reduce the cost of time. (5) For the unmixing methods, the quality and quantity of endmembers have an influence on the results of change detection. Therefore, incorporating a robust endmember extraction method could improve the performance of the change detection procedure. (6) For some methods, the diversity of classes significantly affects the accuracy. This is evident more on the match-based methods.

Disclosure statement

No potential conflict of interest was reported by the authors.

ORCID

Mahdi Hasanlou http://orcid.org/0000-0002-7254-4475
Seyd Teymoor Seydi http://orcid.org/0000-0002-3678-4877

References

Adar, S., G. Notesco, A. Brook, I. Livne, P. Rojik, V. Kopackova, K. Zelenkova, et al. 2011. "Change Detection over Sokolov Open-Pit Mining Area, Czech Republic, Using Multi-Temporal HyMAP Data (2009-2010)." SPIE Remote Sensing, 81800T–81800T, International Society for Optics and Photonics. doi:10.1117/12.898748.

Adar, S., Y. Shkolnisky, and E. Ben Dor. 2012. "New Approach for Spectral Change Detection Assessment Using Multistrip Airborne Hyperspectral Data." 2012 IEEE International Geoscience and Remote Sensing Symposium, 4966–4969, IEEE. doi:10.1109/IGARSS.2012.6352497.

Ahlqvist, O. 2008. "Extending Post-Classification Change Detection Using Semantic Similarity Metrics to Overcome Class Heterogeneity: A Study of 1992 and 2001 US National Land Cover Database Changes." *Remote Sensing of Environment* 112 (3): 1226–1241. doi:10.1016/j.rse.2007.08.012.

Alberti, M., J. M. Marzluff, E. Shulenberger, G. Bradley, C. Ryan, and C. Zumbrunnen. 2003. "Integrating Humans into Ecology: Opportunities and Challenges for Studying Urban Ecosystems." *BioScience* 53 (12): 1169–1179. doi:10.1641/0006-3568(2003)053[1169:IHIEOA]2.0.CO;2.

Bioucas-Dias, J. M., A. Plaza, N. Dobigeon, M. Parente, D. Qian, P. Gader, and J. Chanussot. 2012. "Hyperspectral Unmixing Overview: Geometrical, Statistical, and Sparse Regression-Based Approaches." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 5 (2): 354–379. doi:10.1109/JSTARS.2012.2194696.

Bovolo, F., and L. Bruzzone. 2015. "The Time Variable in Data Fusion: A Change Detection Perspective." *IEEE Geoscience and Remote Sensing Magazine* 3 (3): 8–26. doi:10.1109/MGRS.2015.2443494.

Bovolo, F., S. Marchesi, and L. Bruzzone. 2012. "A Framework for Automatic and Unsupervised Detection of Multiple Changes in Multitemporal Images." *IEEE Transactions on Geoscience and Remote Sensing* 50 (6): 2196–2212. doi:10.1109/TGRS.2011.2171493.

Çeşmeci, D., A. C. Karaca, A. Ertürk, M. K. Güllü, and S. Ertürk. 2014. "Hyperspectral Change Detection by Multi-Band Census Transform." 2014 IEEE Geoscience and Remote Sensing Symposium, 2969–2972, IEEE. doi:10.1109/IGARSS.2014.6947100.

Chen, X. C., V. Kumar, and J. H. Faghmous. 2015. "Online Change Detection Algorithm for Noisy Time-Series: An Application Tonear-Real Time Burned Area Mapping." 2015 IEEE International Conference on Data Mining Workshop (ICDMW) 1536–1537, IEEE. doi:10.1109/ICDMW.2015.237.

Choi, S.-S., S.-H. Cha, and C. C. Tappert. 2010. "A Survey of Binary Similarity and Distance Measures." *Journal of Systemics, Cybernetics and Informatics* 8 (1): 43–48.

Datt, B., T. R. McVicar, T. G. Van Niel, D. L. B. Jupp, and J. S. Pearlman. 2003. "Preprocessing EO-1 Hyperion Hyperspectral Data to Support the Application of Agricultural Indexes." *IEEE Transactions on Geoscience and Remote Sensing* 41 (6): 1246–1259. doi:10.1109/TGRS.2003.813206.

Dekker, R. J., P. B. W. Schwering, K. W. Benoist, S. Pignatti, F. Santini, and O. Friman. 2013. "LWIR Hyperspectral Change Detection for Target Acquisition and Situation Awareness in Urban Areas." SPIE Defense, Security, and Sensing, 874306, International Society for Optics and Photonics. doi:10.1117/12.2015761.

Dhawan, J., and P. Narula. 2015. "An Exploration of Various Change Detection Techniques for Hyperspectral Images." *IPASJ International Journal of Computer Science(IIJCS)* 3 (3): 26–29.

Du, Q., L. Wasson, and R. King. 2005. "Unsupervised Linear Unmixing for Change Detection in Multitemporal Airborne Hyperspectral Imagery." International Workshop on the Analysis of Multi-Temporal Remote Sensing Images, 2005, 136–140, IEEE. doi:10.1109/AMTRSI.2005.1469856.

Eismann, M. T., and J. Meola. 2008. "Hyperspectral Change Detection: Methodology and Challenges." *IGARSS 2008-2008 IEEE International Geoscience and Remote Sensing Symposium* 2: II–605. IEEE. doi:10.1109/IGARSS.2008.4779065.

Eismann, M. T., J. Meola, and R. C. Hardie. 2008. "Hyperspectral Change Detection in the Presenceof Diurnal and Seasonal Variations." *IEEE Transactions on Geoscience and Remote Sensing* 46 (1): 237–249. doi:10.1109/TGRS.2007.907973.

Ertürk, A., and A. Plaza. 2015. "Informative Change Detection by Unmixing for Hyperspectral Images." *IEEE Geoscience and Remote Sensing Letters* 12 (6): 1252–1256. doi:10.1109/LGRS.2015.2390973.

Ertürk, A., M. K. Güllü, D. Çeşmeci, D. Gerçek, and E. Sarp. 2014. "Spatial Resolution Enhancement of Hyperspectral Images Using Unmixing and Binary Particle Swarm Optimization." *IEEE Geoscience and Remote Sensing Letters* 11 (12): 2100–2104. doi:10.1109/LGRS.2014.2320135.

Ertürk, A., M.-D. Iordache, and A. Plaza. 2016. "Sparse Unmixing-Based Change Detection for Multitemporal Hyperspectral Images." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 9 (2): 708–719. doi:10.1109/JSTARS.2015.2477431.

Ertürk, A., M.-D. Iordache, and A. Plaza. 2017. "Sparse Unmixing with Dictionary Pruning for Hyperspectral Change Detection." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 10 (1): 321–330. doi:10.1109/JSTARS.2016.2606514.

Fan, F. 2008. "Digital Change Detection by Post-Classification Comparison of RS Data in Land Use of Guangzhou." *Journal of Computational Information Systems* 4 (2): 1–6.

Ferrato, L.-J., and K. Wayne Forsythe. 2012. "Comparing Hyperspectral and Multispectral Imagery for Land Classification of the Lower Don River." *Toronto, Joulllal of Geography and Geology* 5. doi:10.5539/jgg.v5n1p92.

Foody, G. M. 2002. "Status of Land Cover Classification Accuracy Assessment." *Remote Sensing of Environment* 80 (1): 185–201. doi:10.1016/S0034-4257(01)00295-4.

Fröjse, L. 2011. "Multitemporal Satellite Images for Urban Change Detection." http://www.diva-portal.org/smash/record.jsf?pid=diva2:437205

Ghobadi, Y., B. Pradhan, H. Z. M. Shafri, N. Bin Ahmad, and K. Kabiri. 2015. "Spatio-Temporal Remotely Sensed Data for Analysis of the Shrinkage and Shifting in the Al Hawizeh Wetland." *Environmental Monitoring and Assessment* 187 (1): 1–17. doi:10.1007/s10661-014-4156-0.

Gong, M., Z. Zhou, and M. Jingjing. 2012. "Change Detection in Synthetic Aperture Radar Images Based on Image Fusion and Fuzzy Clustering." *IEEE Transactions on Image Processing* 21 (4): 2141–2151. doi:10.1109/TIP.2011.2170702.

Goodenough, D. G., K. Andrew Dyk, J. S. Olaf Niemann, H. C. Pearlman, T. Han, M. Murdoch, and C. West. 2003. "Processing Hyperion and ALI for Forest Classification." *IEEE Transactions on Geoscience and Remote Sensing* 41 (6): 1321–1331. doi:10.1109/TGRS.2003.813214.

Govender, M., K. Chetty, V. Naiken, and H. Bulcock. 2008. "A Comparison of Satellite Hyperspectral and Multispectral Remote Sensing Imagery for Improved Classification and Mapping of Vegetation." *Water Research Commission (WRC)* 34 (2): 147–154.

Gulati, V., and P.Pal. 2014. "Enhancement of ICA Algorithm Using Mat Lab for Change Detection in Hyper Spectral Images." *International Journal of Education and Science Research Review* 1 (5): 1–9.

Hasanlou, M., and S. T. Seydi. 2016. *Novel Wetland and Water Body Change Detction Using Multitemporal Hyperspectral Imagery*. Oman, Muscat: Springer.

Hussain, M., D. Chen, A. Cheng, H. Wei, and D. Stanley. 2013. "Change Detection from Remotely Sensed Images: From Pixel-Based to Object-Based Approaches." *ISPRS Journal of Photogrammetry and Remote Sensing* 80: 91–106. doi:10.1016/j.isprsjprs.2013.03.006.

İlsever, M., and C. Unsalan. 2012. *Two-Dimensional Change Detection Methods: Remote Sensing Applications*. London, UK: Springer Science & Business Media.

Imaging Spectroscopy. 2018. *Imaging Spectroscopy*. Accessed on http: //www.markelowitz.com/Hyperspectral.html.

Iordache, M.-D., A. Plaza, and B.-D. José. 2010. "On the Use of Spectral Libraries to Perform Sparse Unmixing of Hyperspectral Data." 2010 2nd Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, 1–4, IEEE. doi:10.1109/WHISPERS.2010.5594888.

Júnior, C., A. Osmar, R. F. Guimarães, A. R. Gillespie, N. C. Silva, and R. A. T. Gomes. 2011. "A New Approach to Change Vector Analysis Using Distance and Similarity Measures." *Remote Sensing* 3 (11): 2473–2493. doi:10.3390/rs3112473.

Kamboj, R., and E. M. Ahuja. 2016. "Comparative Analysis of Various Change Detection Techniques for Hyperspectral Images." *Global Journal For Research Analysis* 5 (4): 1–6.

Kawishwar, P. 2007. "Atmospheric Correction Models for Retrievals of Calibrated Spectral Profiles from Hyperion EO-1 Data." *Internatio nal Institute for Geo-in formation Scienc e and Earth Observation*. http://www.itc.nl/library/papers_2007/msc/iirs/kawishwar.pdf

Keshava, N., and J. F. Mustard. 2002. "Spectral Unmixing." *IEEE Signal Processing Magazine* 19 (1): 44–57. doi:10.1109/79.974727.

Khanday, W. A., and K. Kumar. 2016. "Change Detection in Hyper Spectral Images." *Asian Journal of Technology and Management Research (AJTMR) Volume* 6 (02). http://www.ajtmr.com/papers/Vol6Issue2/Vol6Iss2_P8.pdf.

Kumar, V., and K. D. Garg. 2013. "A Valuable Approach for Image Processing and Change Detection on Synthetic Aperture Radar Data." *International Journal of Current Engineering and Technology* 3 (2): 512–516.

Kumari, N., R. Gupta, and E. A. Sharma. 2014. "A Review on Change Detection Methods in Hyper Spectral Image." *International Journal of Research in Information Technology* 2 (4): 764–771.

Lam, A., I. Sato, and Y. Sato. 2012. "Denoising Hyperspectral Images Using Spectral Domain Statistics." Pattern Recognition (ICPR), 2012 21st International Conference on, 477–480, IEEE. http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=6460175.

Lee, S. 2011. "Detecting Wetland Change through Supervised Classification of Landsat Satellite Imagery within the Tunkwa Watershed of British Columbia, Canada." Accessed 20 March 2018. http://www.diva-portal.org/smash/record.jsf?pid=diva2:681571

Li, H., D. Zhang, Y. Zhang, and Y. Xu. 2008. "Research of Image Preprocessing Methods for EO-1 Hyperion Hyperspectral Data in Tidal Flat Area." Geoinformatics 2008 and Joint Conference on

GIS and Built Environment: Classification of Remote Sensing Images, 71471G–71471G, International Society for Optics and Photonics. doi:10.1117/12.813253.

Liu, S. 2015. *Advanced Techniques for Automatic Change Detection in Multitemporal Hyperspectral Images*. Italy: University of Trento. http://eprints-phd.biblio.unitn.it/1393/.

Liu, S., D. Qian, X. Tong, A. Samat, H. Pan, and M. Xiaolong. 2017. "Band Selection-Based Dimensionality Reduction for Change Detection in Multi-Temporal Hyperspectral Images." *Remote Sensing* 9 (10): 1008. doi:10.3390/rs9101008.

Liu, S., L. Bruzzone, F. Bovolo, and D. Peijun. 2016. "Unsupervised Multitemporal Spectral Unmixing for Detecting Multiple Changes in Hyperspectral Images." *IEEE Transactions on Geoscience and Remote Sensing* 54 (5): 2733–2748. doi:10.1109/TGRS.2015.2505183.

Lu, D., L. Guiying, and E. Moran. 2014. "Current Situation and Needs of Change Detection Techniques." *International Journal of Image and Data Fusion* 5 (1): 13–38. doi:10.1080/19479832.2013.868372.

Marpu, P., P. Gamba, and J. A. Benediktsson. 2011. "Hyperspectral Change Detection Using IR-MAD and Feature Reduction." Geoscience and Remote Sensing Symposium (IGARSS), 2011 IEEE International, 98–101, IEEE. doi:10.1109/IGARSS.2011.6048907.

Martin, G., and A. Plaza. 2012. "Spatial-Spectral Preprocessing Prior to Endmember Identification and Unmixing of Remotely Sensed Hyperspectral Data." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 5 (2): 380–395. doi:10.1109/JSTARS.2012.2192472.

Mas, J.-F. 1999. "Monitoring Land-Cover Changes: A Comparison of Change Detection Techniques." *International Journal of Remote Sensing* 20 (1): 139–152. doi:10.1080/014311699213659.

Mousazadeh, R., H. Ghaffarzadeh, J. Nouri, A. Gharagozlou, and M. Farahpour. 2015. "Land Use Change Detection and Impact Assessment in Anzali International Coastal Wetland Using Multi-Temporal Satellite Images." *Environmental Monitoring and Assessment* 187 (12): 1–11. doi:10.1007/s10661-015-4900-0.

Naveena, D. R., and G. W. Jij. 2015. "Change Detection Techniques-A Survey." *International Journal on Computational Sciences & Applications (IJCSA)* 5 (2). doi:10.5121/ijcsa.2015.5205.

Nielsen, A. A. 2007. "The Regularized Iteratively Reweighted MAD Method for Change Detection in Multi-And Hyperspectral Data." *IEEE Transactions on Image Processing* 16 (2): 463–478. doi:10.1109/TIP.2006.888195.

Nielsena, A. A., and A. Müllerb. 2003. "Change Detection by the MAD Method in Hyperspectral Image Data." In *The 3rd EARSeL Workshop on Imaging Spectroscopy*, Germany.

Omo-Irabor, O. O. 2016. "A Comparative Study of Image Classification Algorithms for Landscape Assessment of the Niger Delta Region." *Journal of Geographic Information System* 8 (02): 163. doi:10.4236/jgis.2016.82015.

Otsu, N. 1979. "A Threshold Selection Method from Gray-Level Histograms." *IEEE Transactions on Systems, Man, and Cybernetics* 9 (1): 62–66. doi:10.1109/TSMC.1979.4310076.

Pieper, M., D. Manolakis, E. Truslow, T. Cooley, M. Brueggeman, A. Weisner, and J. Jacobson. 2015b. "Comparison of Hyperspectral Change Detection Algorithms." SPIE Optical Engineering+ Applications, 96110Z–96110Z, International Society for Optics and Photonics. doi:10.1117/12.2188316.

Pieper, M., D. Manolakis, T. Cooley, M. Brueggeman, A. Weisner, and J. Jacobson. 2015a. "New Insights and Practical Considerations in Hyperspectral Change Detection." 2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 4161–4164, IEEE. doi:10.1109/IGARSS.2015.7326742.

Rahman, M. T. 2016. "Detection of Land Use/Land Cover Changes and Urban Sprawl in Al-Khobar, Saudi Arabia: An Analysis of Multi-Temporal Remote Sensing Data." *ISPRS International Journal of Geo-Information* 5 (2): 15. doi:10.3390/ijgi5020015.

Raja, R. A. A., V. Anand, A. Senthil Kumar, S. Maithani, and V. Abhai Kumar. 2013. "Wavelet Based Post Classification Change Detection Technique for Urban Growth Monitoring." *Journal of the Indian Society of Remote Sensing* 41 (1): 35–43. doi:10.1007/s12524-011-0199-7.

Rajeswari, A. V., S. Saritha, and G. Santhosh Kumar. 2014. "Classification Based Land Use/Land Cover Change Detection through Landsat Images." Data Science & Engineering (ICDSE), 2014 International Conference on, 232–237, IEEE. doi:10.1109/ICDSE.2014.6974644.

Resta, S., N. Acito, M. Diani, G. Corsini, T. Opsahl, and T. V. Haavardsholm. 2011. "Detection of Small Changes in Airborne Hyperspectral Imagery: Experimental Results over Urban Areas." Analysis of

Multi-Temporal Remote Sensing Images (Multi-Temp), 2011 6th International Workshop on the, 5–8, IEEE. doi:10.1109/Multi-Temp.2011.6005033.
Rivera, V. O. 2005. “Hyperspectral Change Detection Using Temporal Principal Component Analysis.” Master Thesis, University of Puerto Rico, Spain.
Sallaba, F. 2009. “Potential of A Post-Classification Change Detection Analysis to Identify Land Use and Land Cover Changes: A Case Study in Northern Greece.” *Lunds Universitets Naturgeografiska Institution-Seminarieuppsatser*. http://lup.lub.lu.se/student-papers/record/1887139
Scheffler, D., and P. Karrasch. 2014. “Destriping of Hyperspectral Image Data: An Evaluation of Different Algorithms Using EO-1 Hyperion Data.” *Journal of Applied Remote Sensing* 8 (1): 083645. doi:10.1117/1.JRS.8.083645.
Seydi, S. T., and M. Hasanlou. 2017. “A New Land-Cover Match-Based Change Detection for Hyperspectral Imagery.” *European Journal of Remote Sensing* 50 (1): 517–533. doi:10.1080/22797254.2017.1367963.
Seydi, S. T., and M. Hasanlou. 2018. “Sensitivity Analysis of Pansharpening in Hyperspectral Change Detection.” *Applied Geomatics* 10 (1): 65–75. doi:10.1007/s12518-018-0206-6.
Shah-Hosseini, R., S. Homayouni, and A. Safari. 2015. “A Hybrid Kernel-Based Change Detection Method for Remotely Sensed Data in A Similarity Space.” *Remote Sensing* 7 (10): 12829–12858. doi:10.3390/rs71012829.
Shahid, K. K., K. Staenz, L. Sun, R. Neville, H. P. White, A. Bannari, C. M. Champagne, and R. Hitchcock. 2006. “Preprocessing of EO-1 Hyperion Data.” *Canadian Journal of Remote Sensing* 32 (2): 84–97. doi:10.5589/m06-014.
Shippert, P. 2003. “Introduction to Hyperspectral Image Analysis.” *Online Journal of Space Communication* 3. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.214.8818&rep=rep1&type=pdf.
Sica, Y. V., R. D. Quintana, V. C. Radeloff, and G. I. Gavier-Pizarro. 2016. “Wetland Loss Due to Land Use Change in the Lower Paraná River Delta, Argentina.” *Science of the Total Environment* 568: 967–978. doi:10.1016/j.scitotenv.2016.04.200.
Singh, A. 1989. “Review Article Digital Change Detection Techniques Using Remotely-Sensed Data.” *International Journal of Remote Sensing* 10 (6): 989–1003. doi:10.1080/01431168908903939.
Smith, R. 2012. *Introduction to Hyperspectral Imaging*. Mentor, OH: MicroImages . http://www.microimages.com/documentation/Tutorials/hyprspec.pdf.
Vongsy, K., M. J. Mendenhall, P. M. Hanna, and J. Kaufman. 2009. “Change Detection Using Synthetic Hyperspectral Imagery.” 2009 First Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, 1–4, IEEE. doi:10.1109/WHISPERS.2009.5289016.
Vongsy, K., S. Karimkhan, A. K. Shaw, and D. Wicker. 2007. “Change Detection for Hyperspectral Imagery.” Defense and Security Symposium, 656516, International Society for Optics and Photonics. doi:10.1117/12.723161.
Vongsy, K. M. 2007. “Change Detection Methods for hyperspectral imagery.” *Wright State University*. https://etd.ohiolink.edu/!etd.send_file?accession=wright1184010751&disposition=attachment
Weng, Q. 2011. *Advances in Environmental Remote Sensing: Sensors, Algorithms, and Applications*. 1st ed. Boca Raton, FL: CRC Press. doi:10.1201/b10599-1.
Wu, C., D. Bo, and L. Zhang. 2013. “A Subspace-Based Change Detection Method for Hyperspectral Images.” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 6 (2): 815–830. doi:10.1109/JSTARS.2013.2241396.
Wu, C., L. Zhang, and D. Bo. 2012. “Targeted Change Detection for Stacked Multi-Temporal Hyperspectral Image.” 2012 4th Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS), 1–4, IEEE. doi:10.1109/WHISPERS.2012.6874282.
Xiao Benlin, A., B. Li Fangfang, C. Mao Xingliang, and B. Jin Huazhong. 2008. “Study on Independent Component Analysis Application in Classification and Change Detection of Multispectral Images.” *In The International Archives of The Photogrammetry, Remote Sensing and Spatial Information Sciences, Beijing* XXXVII: 871–875.
Yang, Y., and Z. Yan. 2016. “Monitoring and Analyzing of Poyang Lake Wetland Land Use Change Based on RS and GIS.” Geo-Informatics in Resource Management and Sustainable Ecosystem, 213–221, Springer. doi:10.1007/978-3-662-49155-3_21.

Yang, Z., and R. Mueller. 2007. “Spatial-Spectral Cross-Correlation for Change Detection—A Case Study for Citrus Coverage Change Detection.” ASPRS 2007 Annual Conference. http://www.asprs.org/wp-content/uploads/2011/01/0081.pdf.

Yuan, Y., H. Lv, and X. Lu. 2015. “Semi-Supervised Change Detection Method for Multi-Temporal Hyperspectral Images.” *Neurocomputing* 148: 363–375. doi:10.1016/j.neucom.2014.06.024.